



BizFinBench.v2: A Unified Dual-Mode Bilingual Benchmark for Expert-Level Financial Capability Alignment

Xin Guo^{1,2,*}, Rongjunchen Zhang^{1,*♣}, Guilong Lu¹, Xuntao Guo¹,
Shuai Jia¹, Zhi Yang², Liwen Zhang^{2,♣}

¹HiThink Research, ²Shanghai University of Finance and Economics

zhangrongjunchen@myhexin.com, zhang.liwen@shufe.edu.cn

Abstract

Large language models have undergone rapid evolution, emerging as a pivotal technology for intelligence in financial operations. However, existing benchmarks are often constrained by pitfalls such as reliance on simulated or general-purpose samples and a focus on singular, offline static scenarios. Consequently, they fail to align with the requirements for authenticity and real-time responsiveness in financial services, leading to a significant discrepancy between benchmark performance and actual operational efficacy. To address this, we introduce BizFinBench.v2, the first large-scale evaluation benchmark grounded in authentic business data from both Chinese and U.S. equity markets, integrating online assessment. We performed clustering analysis on authentic user queries from financial platforms, resulting in eight fundamental tasks and two online tasks across four core business scenarios, totaling 29,578 expert-level Q&A pairs. Experimental results demonstrate that ChatGPT-5 achieves a prominent 61.5% accuracy in main tasks, though a substantial gap relative to financial experts persists; in online tasks, DeepSeek-R1 outperforms all other commercial LLMs. Error analysis further identifies the specific capability deficiencies of existing models within practical financial business contexts. BizFinBench.v2 transcends the limitations of current benchmarks, achieving a business-level deconstruction of LLM financial capabilities and providing a precise basis for evaluating efficacy in the widespread deployment of LLMs within the financial domain. The data and code are available at <https://github.com/HiThink-Research/BizFinBench.v2>.

1 Introduction

Large Language Models (LLMs) have developed rapidly in recent years, and their application bound-

*These authors contributed equally to this work.

♣Corresponding authors.

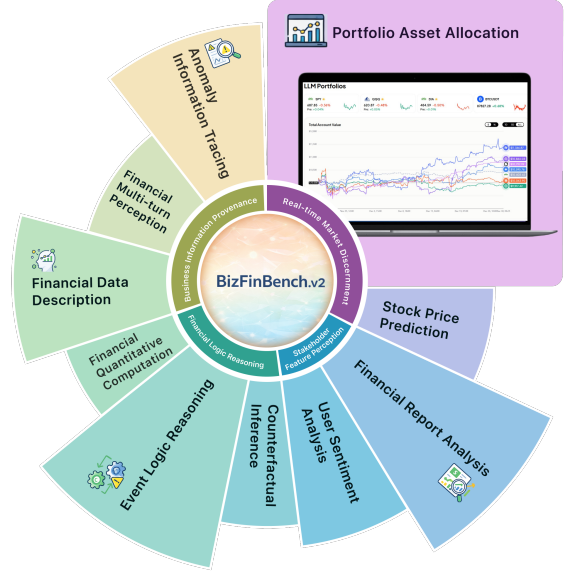


Figure 1: BizFinBench.v2 comprises eight foundational tasks and two online tasks distributed across four major scenarios. The top-right corner displays a real-time screenshot of the Portfolio Asset Allocation task.

aries in the financial sector continue to expand, making them an important technological direction for promoting the intelligent upgrade of financial services (Chen et al., 2024; Liu et al., 2025a; Zhang et al., 2023b; Lu et al., 2024). However, as the scope of LLMs’ application in the financial sector becomes increasingly broad, the discrepancy between model performance under existing evaluation paradigms and their actual performance in real financial business scenarios has become increasingly prominent, directly leading to a conflict between real financial business needs and the limitations of various evaluation benchmarks.

The core characteristics of financial scenarios lie in authenticity and online capability, yet the vast majority of current financial benchmarks are undermined by two fundamental limitations:

(1) Disconnected from Real-world Business

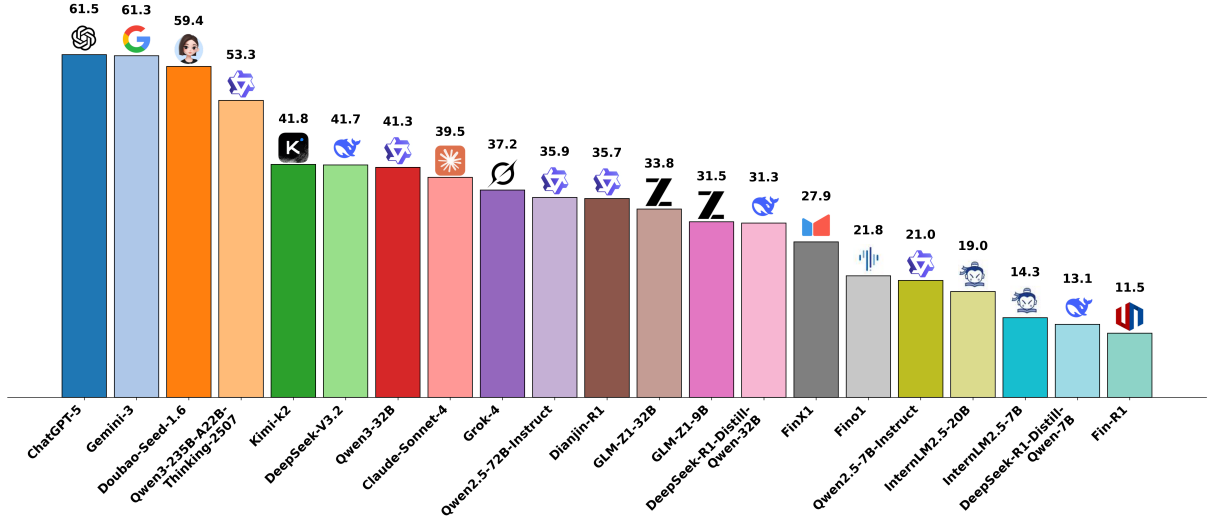


Figure 2: We have ranked the performance of the LLMs participating in the evaluation under the zero-shot setting, and these results reflect their authentic practical business capabilities.

Scenarios: Most existing studies rely on simulated samples or general datasets that are disconnected from actual business operations. These datasets generally exhibit low difficulty and are decoupled from the requirements of real-world financial business, as they lack the core logic of business workflows and fail to reflect the practical challenges inherent in the financial domain. (Zhu et al., 2024; Jiang et al., 2025).

(2) Focusing on Purely Static Offline Tasks:

Existing benchmarks universally focus on purely offline static tasks (Xie et al., 2024), lacking coverage of online tasks in financial scenarios. This prevents the support for evaluating model performance in online business scenarios such as real-time market data push and dynamic risk monitoring, thus being disconnected from the core requirement of online capability in financial services (Tang et al., 2025). These two problems result in a disconnect between existing benchmarks and real-world applications, leading to a significant gap between LLM benchmark test performance and actual business performance.

To bridge the gap between current LLM evaluation performance and actual operational efficacy, we introduce BizFinBench.v2, the first large-scale financial business evaluation benchmark that integrates authentic business data from both Chinese and U.S. equity markets and features a dual-track evaluation of "Core Business Capabilities + Online Performance." BizFinBench.v2 comprises 29,578 QA pairs based on real business data. Through

the clustering of massive user queries from authentic business platforms and rigorous quality control, we have categorized four core business scenarios: Business Information Provenance, Financial Logic Reasoning, Stakeholder Feature Perception, and Real-time Market Discernment. These encompass eight fundamental tasks, including Anomaly Information Tracing, Financial Multi-turn Perception, Financial Data Description, Financial Quantitative Computation, Event Logic Reasoning, Counterfactual Inference, User Sentiment Analysis and Financial Report Analysis, along with two online tasks: Stock Price Prediction and Portfolio Asset Allocation. The detailed architecture is illustrated in Figure 1.

Unlike existing benchmarks, BizFinBench.v2 not only enables a precise and objective quantitative evaluation of the actual efficacy of LLMs in authentic financial scenarios but also identifies core capability shortcomings in financial business applications. It achieves a comprehensive deconstruction of LLM capabilities from a business perspective, thereby providing a reliable performance reference for the deployment of LLMs in the financial domain.

Our contributions are threefold:

(1) **Authentic Business Data Driven:** We propose the first large-scale financial evaluation benchmark driven by authentic data. Developed upon a core foundation of real-world business data from Chinese and U.S. equity markets, BizFinBench.v2 transcends the limitations inherent in existing eval-

uation frameworks, specifically their disconnection from practical business scenarios and their reliance on low-complexity, general-purpose, or synthetic data.

(2) **Innovative Dual-Track Evaluation Framework:** We propose a dual-track evaluation framework encompassing "Core Business Capabilities + Online Performance," thereby addressing the deficiencies in application scenario coverage inherent in traditional offline static assessments.

(3) **Expert-Informed Business Error Analysis:** We provide an error analysis of the actual performance of LLMs from the business perspective of financial experts, offering targeted optimization directions for broader business adaptation.

The organization of this paper is as follows: Section 2 deliberates on the current status of various operations and benchmarks in the financial domain, identifying the corresponding requirements and the limitations of existing research. Section 3 delineates the construction, quality control, and detailed statistics of BizFinBench.v2. Sections 4 and Section 5 primarily describe the experimental setup, as well as the main experimental results and analysis. Section 6 presents an error analysis conducted from an expert business perspective. Finally, Section 7 concludes the study and proposes directions for future research.

2 Related Work

2.1 Financial Business Analysis

In financial business scenarios, LLM-based tasks such as text understanding (Wilson et al., 2024), sentiment analysis (Delgadillo et al., 2024; Zhang et al., 2023a), time-series forecasting (Li et al., 2024a,b; Wang et al., 2024; Mai, 2024), and wealth management & investment (Yu et al., 2024; Wang et al., 2025; Yu et al., 2025; Yang et al., 2023) serve as foundational components for building universal financial business capabilities. They provide crucial technical support for judging market trends and mitigating potential risks. However, the financial sector’s demand for timeliness means standalone offline tasks cannot fully evaluate LLMs’ performance in real-world business contexts—especially scenarios closely tied to dynamic financial markets, such as stock price fluctuation prediction (Yu and Yan, 2020), real-time response to market hot topics and public opinion (Mao et al., 2025), intraday trading signal interpretation (Cervelló-Royo et al., 2015), and instant investment decisions (Suresh,

2024). Nevertheless, most current research on LLMs in finance focuses on enhancing offline question-answering quality and optimizing reasoning accuracy, verifying model performance through static datasets (Liu et al., 2025b; Zhu et al., 2025). Few studies test LLMs’ capabilities in processing real-time market data and generating instantaneous decisions from the perspective of financial businesses’ online and real-time needs. This gap results in a mismatch between LLM performance and the core demands of financial markets, leaving significant room for future exploration.

2.2 Financial Benchmark Analysis

Mainstream evaluation benchmarks in the current financial domain have achieved multi-dimensional coverage for assessing the foundational capabilities of LLMs. Benchmarks such as CFLUE (Zhu et al., 2024), FinEval (Guo et al., 2025b), and CGCE (Zhang et al., 2023c) focus on basic financial knowledge and text comprehension, encompassing tasks like text classification and policy interpretation. Others including FinQA (Chen et al., 2021), ConvFinQA (Chen et al., 2022), FinanceReasoning (Tang et al., 2025), and FinBen (Xie et al., 2024) emphasize financial numerical reasoning and complex decision-making, simulating quantitative analysis scenarios such as financial report interpretation. Meanwhile, benchmarks like DISC-FINSFT (Chen et al., 2023), PIXIU (Xie et al., 2023), FinMaster (Jiang et al., 2025) and CFBenchmark (Lei et al., 2023) prioritize instruction following and tool adaptation. However, these benchmarks are generally focused on verifying performance in single, offline, and static tasks, failing to accommodate the real-time demands inherent in financial services. Furthermore, the construction of the relevant data is mostly derived from simulated synthesis or manual rewriting, making it difficult to align with real-world financial business requirements. Simultaneously, there is a lack of in-depth discussion on the consistency between evaluated performance and the LLM’s actual financial application capability, leading to a divergence between the practical guidance provided by the benchmarks and the core demands of the financial market. Detailed comparison with existing benchmarks can be found in Table 1.

Table 1: In-depth multi-dimensional comparison between BizFinBench.v2 and existing benchmarks from a real-business perspective. The abbreviations and their core meanings are as follows: Cross-Lingual (CL), On-line Testing (OT), Real Business Data (RBD).

Benchmarks	CL	OT	RBD
FinEval (Guo et al., 2025b)	✗	✗	✗
CFLUE (Zhu et al., 2024)	✓	✗	✗
FinQA (Chen et al., 2021)	✗	✗	✗
ConvFinQA (Chen et al., 2022)	✗	✗	✗
FinMaster (Jiang et al., 2025)	✗	✗	✗
FinBen (Xie et al., 2024)	✓	✗	✗
FinanceMath (Zhao et al., 2024)	✗	✗	✗
FinanceReasoning (Tang et al., 2025)	✗	✗	✗
BizFinBench.v2	✓	✓	✓

3 BizFinBench.v2 Benchmark

3.1 Overview

As the first large-scale financial evaluation benchmark to cover authentic business data from both Chinese and U.S. equity markets while integrating offline evaluation and online testing, BizFinBench.v2 aims to bridge the gap between the performance of LLMs in the financial domain and their actual business capabilities. All data within the benchmark is derived from real financial business platforms and has undergone rigorous compliance review and quality control; the dataset will be released as open-source for research purposes in the future.

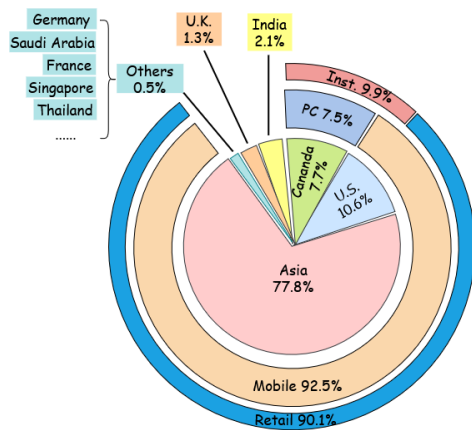


Figure 3: The platform’s user structure is primarily analyzed through user type, device distribution, and national distribution; here, Inst. serves as an abbreviation for institutional users.

3.2 Task Construction

In the process of task construction, BizFinBench.v2 anchors itself in the core functions of authentic financial business platforms and is guided by the

actual demands of key market participants. It covers the diverse needs of individual users, including novices, short-term investors, and high-net-worth individuals, while accounting for the critical requirements of mainstream financial institutions such as banks and securities firms. The platform user distribution is shown in Figure 3.

Business Information Provenance (BIP) focuses on frontline scenarios, requiring LLMs to resolve various business inquiries through direct interaction. This includes real-world perturbations such as erroneous input, irrelevant information, or typos, necessitating robust discernment capabilities from the models. Specifically, the Anomaly Information Tracing (AIT) task addresses user inquiries regarding the root causes of market anomalies, such as significant fluctuations in watchlist stocks; it requires LLMs to filter key clues from massive multi-dimensional information, exclude interference, and precisely identify the core events driving stock or market volatility. The Financial Multi-turn Perception (FMP) task requires LLMs to respond to continuous inquiries within long-context interactions, integrating historical user queries to provide sustained service. The Financial Data Description (FDD) task demands that LLMs judge the authenticity and accuracy of specific data within various information sources to provide precise conclusions.

The Financial Logical Reasoning (FLR) scenario serves as both a foundational element of Business Information Provenance, requiring LLMs to perform rigorous financial reasoning based on market data returned by various interfaces. Within this category, the Financial Quantitative Computation (FQC) task requires LLMs complete accurate indicator calculations based on complex financial data. The Event Logical Reasoning (ELR) task requires LLMs to sort macro or micro market events based on chronological order or causal logic. The Counterfactual Inference (CI) task requires LLMs to simulate financial experts by conducting reasonable logical analysis and judgment based on user-proposed hypotheses, outputting reliable and valid conclusions.

The Stakeholder Feature Perception (SFP) scenario aims to provide users with in-depth analysis and summaries of the market or industry. The User Sentiment Analysis (SA) task requires LLMs to classify and evaluate overall user sentiment through the quantitative analysis of various user information and behaviors, providing support for customized services and highly relevant product

recommendations. The Financial Report Analysis (FRA) task requires LLMs to conduct quantitative analysis by combining technical, fundamental, and news-based periodic report information to determine enterprise industry rankings, helping users intuitively grasp corporate qualifications and future prospects. This task is limited to the Chinese market.

Finally, the Real-time Market Discernment (RMD) scenario covers two types of online tasks of high interest to users: Stock Price Prediction (SPP) and Portfolio Asset Allocation (PAA). It should be noted that the real-time nature of online tasks requires all relevant data to be derived from the actual equity market, and theoretically, the number of questions is unlimited. However, due to the static nature of a research paper, the cutoff time for all tasks is set to December 24, 2025. The Stock Price Prediction task addresses user needs for predicting stocks in their watchlists, requiring LLMs to predict daily closing prices by integrating historical and real-time stock prices, technical indicators, and relevant news. The Portfolio Asset Allocation task presents a higher degree of difficulty. We have constructed an LLM investment simulation system for this task that restores real-world trading rules, including transaction fees, latency, and slippage, on top of real-time data access. Through systematic prompt engineering and various data and tool interfaces, LLMs can perform asset allocation in a real market environment using simulated funds. Decisions are generated hourly during the day, allowing for either trading or skipping, with each decision cycle covering the full process from reasoning and analysis to order submission and trade execution. We will subsequently open-source this LLM investment system to allow all researchers interested in the financial field to integrate different open-source or proprietary LLMs for in-depth study.

For a more detailed introduction to the above content, please refer to Appendix A.

3.3 Quality Control

For tasks in offline tasks, the quality control process strictly adopts a three-level progressive mechanism of "Platform Clustering and Desensitization - Frontline Staff Review - Expert Team Cross-Validation" to ensure the validity and compliance of the evaluation data. First, all raw data undergo clustering through the platform's multi-dimensional algorithms to automatically aggregate and categorize different task types. Simultaneously, data desensiti-

zation is performed (including user PII or sensitive corporate data), and a preliminary review is conducted in alignment with financial industry compliance requirements to eliminate content that clearly violates regulatory standards. Second, ten frontline business professionals with over five years of experience manually screen the data samples one by one to filter out invalid Q&A pairs, duplicate records, and samples with abnormal expressions, while retaining specific real-world perturbations to ensure each Q&A pair fully reflects authentic business scenarios. Finally, we invited six senior financial experts, each with over a decade of industry experience from three different professional teams within the Financial Research Institute, to conduct cross-validation in three groups. Ultimately, through this rigorous three-level screening mechanism, high-quality Q&A pairs are formed that balance authenticity, business alignment, and information security.

As for online tasks, the financial experts mentioned above strictly defined the structured data required for the Stock Price Prediction task, the specific equity market configurations for the Portfolio Asset Allocation task, and the system prompts, ensuring full compliance with the requirements of authentic business scenarios.

It should be noted that for offline tasks, due to the protection of financial business privacy, the specific evaluation rubrics and quantitative standards employed during the quality review process cannot be disclosed to the public. In contrast, for online tasks, as real-time public equity market data is utilized and the configurations are designed to replicate the actual market environment, the details of these settings and the system prompts will be made open-source.

3.4 Statistics

BizFinBench.v2 covers the two major equity markets (China and the U.S.), with all data drawn from authentic user demands within financial business scenarios. This has resulted in four core scenarios: BIP, FLR, SFP and RMD. These comprise eight offline tasks and two online tasks, totaling 29,578 questions derived from real-world financial service requests. Specifically, the AIT, ELR, SA, and CI tasks consist of 4,000 questions each, with average input token counts of 8,679, 437, 3,326, and 5,510, respectively. FQC and FRA each contain 2,000 questions, with average input token counts of 1,984 and 19,681. FMP and FDD consist of 3,741 and 3,837 questions, with average input token counts

of 10,361 and 3,577, respectively. The SPP task, based on 100 popular stocks, accumulated a total of 4,000 questions with an average input token count of 5,510. Regarding the PAA task, due to its unique nature as an online task, we do not include the number of transactions in the dataset; instead, we evaluate the actual performance of the LLMs by comparing final metrics based on cumulative results. More intuitive statistical results can be found in Table 5 within Appendix A.

4 Experiment Setting

4.1 Evaluated Methods

All questions in BizFinBench.v2 are verifiable open-ended questions. Except for the Portfolio Asset Allocation task, all other tasks utilize Accuracy as the primary metric. We provide additional clarifications here: (1) For the User Sentiment Analysis and Stock Price Prediction tasks, we draw upon the concept of conformal prediction algorithms (Shafer and Vovk, 2008) used in machine learning. Models are required to construct prediction intervals based on preset business tolerance levels; a prediction is judged valid by checking whether the standard answer falls within the model-provided interval. Relevant results are labeled in the `interval_boxed{}` format for verification. Here, the business tolerance for the sentiment analysis task is set at 10%, and for the stock price prediction task, it is set at 1%. (2) Metrics for the Portfolio Asset Allocation task primarily consist of practical business indicators such as cumulative return, Sharpe ratio, and maximum drawdown; therefore, the models evaluated for this task (mainly six proprietary commercial models) and their results are independent of other tasks. In the experimental results section, we present the average ranking of models across different metrics as the final result. Given the variations in input length across different tasks and the constraints of model context windows, the primary evaluation method is standardized as zero-shot. For further details regarding experimental settings, please refer to Appendix B.1.

4.2 Evaluated Models

We evaluated 21 LLMs, where proprietary models were accessed via their respective APIs and open-source models were deployed locally. All inference tasks were executed on an 8×NVIDIA H100 GPU cluster. For additional details regarding the models, refer to Appendix B.2. We additionally invited

two financial experts who were not involved in the data construction process, to participate in the task evaluation of BizFinBench.v2 to provide a performance comparison with the LLMs.

In the realm of proprietary LLMs, we evaluated 6 leading, high-performance models, including ChatGPT-5 (OpenAI, 2023), Gemini-3 (Team et al., 2023), Claude-Sonnet-4 (Anthropic, 2025), Grok-4 (xAI, 2025), Doubao-Seed-1.6 (Bytedance, 2025) and Kimi-k2 (Moonshot, 2025). For open-source models, we evaluated 11 models from several mainstream LLMs, including Qwen2.5-7B/72B-Instruct, Qwen3-32B and Qwen3-235B-A22B-Thinking-2507 (Bai et al., 2023), InternLM2.5-7B/20B (Team, 2023), GLM-Z1-9B/32B (GLM et al., 2024), DeepSeek-V3.2 and Deepseek-R1-Distill-Qwen-7B/32B (Guo et al., 2025a). In financial domain, we evaluated 4 representative LLMs tailored for the financial tasks, including Fin-R1 (Liu et al., 2025b), Dianjin-R1 (Zhu et al., 2025), FinX1 (Team, 2024) and Fino1 (Qian et al., 2025). In the PAA task, we have included Qwen3-Max as an additional model for evaluation.

5 Main Results

The complete model ranking results are shown in Figure 2. Table 2 presents the quantitative performance of all evaluated models across various tasks. From an overall performance perspective, ChatGPT-5 ranks first among all participating models with an average accuracy of 61.5%, highlighting its comprehensive competitive advantage in financial scenarios. Furthermore, proprietary models such as Gemini-3 and Doubao-Seed-1.6 also demonstrate excellent performance, ranking within the top three across multiple tasks. Within the category of open-source models, Qwen3-235B-A22B-Thinking-2507 emerges as the top-performing model with an average accuracy of 53.3%. In contrast, the leading financial model, Dianjin-R1, achieves an average accuracy of only 35.7%, trailing Qwen3-32B by a margin of 5.6%. We analyze this discrepancy from two perspectives: first, the training data for financial-specific models is primarily derived from open-source financial datasets centered on financial knowledge and simulated scenarios, which fail to map the complex characteristics of real-world financial business environments; second, although Dianjin-R1 incorporates customer service Q&A data, its business coverage remains narrow, mak-

Table 2: LLM performance in zero-shot setting on BizFinBench.v2 (%). The results are color-coded to indicate the top three performers in each task: **gold** represents the best-performing model, **silver** represents the second-best result, and **bronze** represents the third-best performance.. The meanings of the abbreviations are as follows: AIT(Anomaly Information Tracing), FMP(Financial Multi-turn Perception), FDD(Financial Data Description), FQC(Financial Quantitative Computation), ELR(Event Logic Reasoning), CI(Counterfactual Inference), SA(User Sentiment Analysis), FRA(Financial Report Analysis), SPP(Stock Price Prediction).

Model	Size	AIT	FMP	FDD	FQC	ELR	CI	SA	FRA	SPP	Average
Proprietary LLMs											
ChatGPT-5	unknown	54.2	90.8	68.3	89.2	62.0	83.9	18.8	54.1	32.1	61.5
Gemini-3	unknown	64.8	87.0	69.7	85.8	69.5	82.2	7.4	50.8	34.9	61.3
Doubao-Seed-1.6	unknown	62.6	90.2	63.8	78.2	61.2	78.7	22.7	46.3	31.1	59.4
Kimi-k2	unknown	55.2	80.9	22.4	62.2	44.6	15.4	20.1	45.0	30.5	41.8
Claude-Sonnet-4	unknown	54.8	79.9	28.4	29.8	44.8	23.4	17.3	47.7	29.1	39.5
Grok-4	unknown	61.8	86.6	37.4	9.3	42.1	4.7	17.8	45.2	30.3	37.2
Open-source General LLMs											
Qwen3-235B-A22B-Thinking-2507	235B	49.3	87.9	68.0	76.0	50.6	72.2	16.4	22.5	36.9	53.3
DeepSeek-V3.2	685B	60.2	87.4	42.3	22.0	49.4	8.3	23.5	50.2	32.8	41.7
Qwen3-32B	32B	54.3	80.9	40.0	48.4	42.0	47.2	13.4	40.5	5.1	41.3
Qwen2.5-72B-Instruct	72B	61.0	78.2	26.5	19.8	39.9	20.7	21.6	47.0	8.2	35.9
GLM-Z1-32B	32B	49.6	66.9	45.6	34.4	31.6	31.3	1.2	36.8	6.8	33.8
GLM-Z1-9B	9B	46.5	69.2	40.4	26.9	35.2	25.2	0.4	36.7	3.4	31.5
DeepSeek-R1-Distill-Qwen-32B	32B	52.1	75.4	20.5	21.0	35.9	24.2	18.4	27.3	6.8	31.3
Qwen2.5-7B-Instruct	7B	35.2	43.2	33.8	0.8	20.9	1.0	23.3	28.3	2.8	21.0
InternLM2.5-20B	20B	32.1	41.1	32.0	0.2	27.8	0.7	3.7	32.5	0.6	19.0
InternLM2.5-7B	7B	28.3	14.8	30.6	0.5	18.9	0.3	6.7	28.5	0.0	14.3
DeepSeek-R1-Distill-Qwen-7B	7B	16.6	30.9	17.1	3.5	10.3	6.2	19.1	13.0	1.0	13.1
Open-source Financial LLMs											
Dianjin-R1	32B	54.2	70.7	40.9	25.3	45.8	22.0	6.7	46.0	9.9	35.7
FinX1	70B	47.9	73.0	29.5	14.3	31.5	11.6	5.3	34.7	3.6	27.9
Finol	14B	27.5	38.6	24.3	14.1	14.6	22.1	11.1	35.3	8.6	21.8
Fin-R1	7B	21.2	29.1	0.5	1.5	9.8	3.2	10.9	24.5	2.9	11.5
Financial Experts	–	92.6	98.0	94.5	100	91.7	100	57.9	96.0	32.3	84.8

Table 3: Comparison of key metrics in Portfolio Asset Allocation tasks. The primary indicators include Total Return (TR), Profit Factor (PF), Sharpe Ratio (SR), Maximum Drawdown (MD), and Total Assets (TA).

Model	TR	PF	SR	MD	TA
DeepSeek-R1	+13.46%	2.3	1.8	-8%	11346
Qwen3-Max	+8.43%	1.8	1.4	-10%	10843
Grok-4	+7.97%	1.4	1.1	-1%	10797
Gemini-3	+5.88%	1.4	1.1	-6%	10588
Claude-Sonnet-4	+0.10%	1.1	0.8	-12%	10010
ChatGPT-5	-0.92%	0.9	0.5	-6%	9908
SPY	–	–	–	–	10474

ing it difficult to adapt to more volatile and long-context practical business scenarios.

From a task perspective, models with larger parameter scales perform better on tasks that demand high data precision, such as FDD, FQC, and CI. Notably, the proprietary models Claude-Sonnet-4 and Grok-4 show a significant gap compared to other proprietary models in these tasks, indicating that their computational capabilities require further enhancement to adapt to the application requirements of the financial domain. The results of the

SA task reflect the difficulty of models in executing subjective analysis tasks; even the top-performing DeepSeek-V3.2 achieved an average accuracy of only 23.5%, which is far below that of financial experts. Furthermore, in the SPP task, the best-performing Qwen3-235B-A22B-2507 reached an accuracy of only 36.9%, confirming that current LLMs are still unable to meet the high demands of online tasks for precise stock movement prediction based on multi-dimensional information. Table 3 presents the key metrics of six mainstream proprietary models in the asset allocation task, revealing significant differences in their real-world investment capabilities. DeepSeek-R1 demonstrates the most outstanding performance, leading in metrics such as total return, profit-loss ratio, and Sharpe ratio, while maintaining a maximum drawdown of -8%, reflecting its superior risk-reward balancing capability. In contrast, ChatGPT-5 and Claude Sonnet-4 failed to outperform the market benchmark (SPY); specifically, ChatGPT-5 experienced a slight loss, with both its profit-loss ratio and Sharpe ratio falling below 1, suggesting that its strategy ad-

justment capabilities need improvement. Notably, although Grok-4’s return is moderate, its maximum drawdown is only -1%, demonstrating exceptional risk control.

Viewed comprehensively, in foundational capability testing, LLMs remain far below the level of financial experts (84.8%), with a significant gap existing between the two; however, in online tasks, mainstream commercial proprietary models have demonstrated a certain degree of potential. This implies that there is still substantial room for improvement regarding the adaptation capabilities and practical potential of current LLMs in real-world financial business scenarios. Subsequent optimization efforts should focus on dimensions such as training data quality and scenario adaptability.

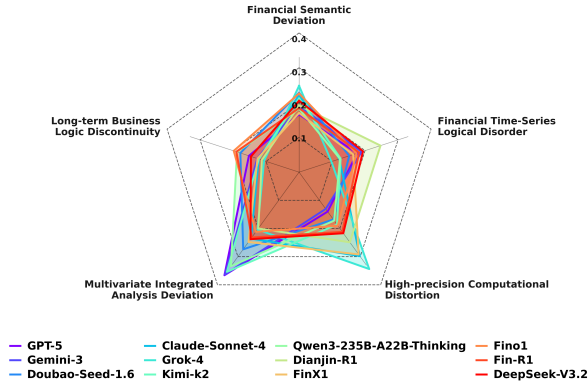


Figure 4: We selected representative LLMs for error analysis and summarized five typical dilemmas faced by LLMs in actual business scenarios

6 Error Analysis

To further investigate the performance deficiencies of LLMs in financial business scenarios, we conducted a targeted analysis on a random sample of 20% of the incorrect responses from all evaluated models. By deconstructing model issues through the lens of business practice, we have identified five typical business dilemmas prevalent across current LLMs in financial tasks: Financial Semantic Deviation, Long-term Business Logic Discontinuity, Multivariate Integrated Analysis Deviation (MIAD), High-precision Computational Distortion, and Financial Time-Series Logical Disorder. According to the distribution of error types in Figure 4, the proportion of error types for the vast majority of models exhibits similar characteristics, reflecting the common challenges currently faced by LLMs in financial business adaptation; how-

ever, certain models also show significant differences. Specifically, ChatGPT, Gemini-3, Doubao-Seed-1.6, and Qwen3-235B-A22B-Thinking have the highest proportion of errors in MIAD, indicating that these models possess distinct performance bottlenecks in information integration and analysis that require targeted optimization. Furthermore, higher error proportions suggest that Grok-4 and Claude-Sonnet-4 need to improve their high-precision computational capabilities, while other models with relatively smaller parameter scales require full-process business optimization to better adapt to relevant financial scenarios. For specific error examples and detailed descriptions of each business dilemma, please refer to Appendix D.

7 Conclusion

In this study, we innovatively present BizFinBench.v2, the first evaluation benchmark constructed based on authentic user data from real-world platforms. By encompassing offline and online tasks across Chinese and U.S. equity markets, it effectively addresses the dual limitations of existing financial LLM research and provides a scientific evaluation framework tailored to practical business applications. We conducted a comprehensive multi-dimensional assessment and quantitative analysis of 21 mainstream open-source and proprietary LLMs, with primary results indicating that ChatGPT-5 achieved the best overall performance, demonstrating superior practical capabilities across various financial tasks. Qwen3-235B-A22B-Thinking-2507 ranked among the top of open-source models, though an 8.2% performance gap remains compared to ChatGPT-5, while Dianjin-R1 led among domain-specific financial models despite a more pronounced gap behind the top two. In the Portfolio Asset Allocation tasks, DeepSeek-R1 stood out among all evaluated commercial models by achieving a 13.46% total return while maintaining a relatively favorable maximum drawdown of -8%, showcasing an excellent balance between profitability and risk control. Furthermore, through deep error analysis of model outputs, we distilled five typical dilemmas faced by LLMs in financial business scenarios, which reflect the unique challenges and value of BizFinBench.v2 in assessing business adaptability. As a financial benchmark born from real user data, BizFinBench.v2 provides an objective and reproducible quantitative standard for measuring the real-world business performance

of LLMs, and we expect it to drive the targeted optimization and iterative upgrading of LLMs in the financial domain, providing critical technical support for the large-scale and reliable deployment of LLMs in financial settings.

Limitations

While BizFinBench.v2 has achieved significant progress in the field of LLM evaluation driven by real-world business data, its application still has three limitations that subsequent research can further improve: first, in terms of data coverage, although the Q&A pairs in BizFinBench.v2 are primarily derived from user data of real business platforms, the core acquisition method relies on platform data clustering, which may result in the exclusion of niche user query types with low proportions, necessitating a further expansion of data source scope in the future; second, regarding the dimension of task design, the current online evaluation tasks only focus on the two scenarios with the highest user demand—stock price prediction and asset allocation tasks directly related to investment—meaning subsequent research could extend the evaluation scenarios to a broader range of online financial business links, such as personalized stock recommendations and account opening guidance; and finally, at the level of evaluation paradigms, since the Q&A data of BizFinBench.v2 are all derived from real platforms and the input text lengths are generally long, the current evaluation mainly concentrates on two paradigms, Zero-Shot evaluation and Chain-of-Thought evaluation, requiring the addition of Few-Shot evaluation experiments in the future.

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References

Anthropic. 2025. Claude4.5. <https://www.anthropic.com/>.

Jinze Bai, Shuai Bai, and et al. Yunfei Chu. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609*.

Bytedance. 2025. doubao. <https://seed.bytedance.com>.

Roberto Cervelló-Royo, Francisco Guijarro, and Karolina Michniuk. 2015. Stock market trading rule based on pattern recognition and technical analysis: Forecasting the djia index with intraday data. *Expert systems with Applications*, 42(14):5963–5975.

Wei Chen, Qiushi Wang, Zefei Long, Xianyin Zhang, Zhongtian Lu, Bingxuan Li, Siyuan Wang, Jiarong Xu, Xiang Bai, Xuanjing Huang, and 1 others. 2023. Disc-finllm: A chinese financial large language model based on multiple experts fine-tuning. *arXiv preprint arXiv:2310.15205*.

Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan R Routledge, and 1 others. 2021. Finqa: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711.

Zhiyu Chen, Shiyang Li, Charese Smiley, Zhiqiang Ma, Sameena Shah, and William Yang Wang. 2022. Convinqa: Exploring the chain of numerical reasoning in conversational finance question answering. *arXiv preprint arXiv:2210.03849*.

Zhiyu Zoey Chen, Jing Ma, Xinlu Zhang, Nan Hao, An Yan, Armineh Nourbakhsh, Xianjun Yang, Julian McAuley, Linda Petzold, and William Yang Wang. 2024. A survey on large language models for critical societal domains: Finance, healthcare, and law. *arXiv preprint arXiv:2405.01769*.

Josiel Delgadillo, Johnson Kinyua, and Charles Mutigwe. 2024. Finsosent: Advancing financial market sentiment analysis through pretrained large language models. *Big Data and Cognitive Computing*, 8(8):87.

Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Diego Rojas, Guanyu Feng, Hanlin Zhao, Hanyu Lai, Hao Yu, Hongning Wang, Jiadai Sun, Jiajie Zhang, Jiale Cheng, Jiayi Gui, Jie Tang, Jing Zhang, Juanzi Li, and 37 others. 2024. *Chatglm: A family of large language models from glm-130b to glm-4 all tools*. *Preprint*, arXiv:2406.12793.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, and 1 others. 2025a. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Xin Guo, Haotian Xia, Zhaowei Liu, Hanyang Cao, Zhi Yang, Zhiqiang Liu, Sizhe Wang, Jinyi Niu, Chuqi Wang, Yanhui Wang, and 1 others. 2025b. Fineval: A chinese financial domain knowledge evaluation benchmark for large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6258–6292.

- Junzhe Jiang, Chang Yang, Aixin Cui, Sihan Jin, Ruiyu Wang, Bo Li, Xiao Huang, Dongning Sun, and Xin-run Wang. 2025. Finmaster: A holistic benchmark for mastering full-pipeline financial workflows with llms. *arXiv preprint arXiv:2505.13533*.
- Yang Lei, Jiangtong Li, Dawei Cheng, Zhijun Ding, and Changjun Jiang. 2023. Cfbenchmark: Chinese financial assistant benchmark for large language model. *arXiv preprint arXiv:2311.05812*.
- Xiang Li, Zhenyu Li, Chen Shi, Yong Xu, Qing Du, Mingkui Tan, and Jun Huang. 2024a. Alphafin: Benchmarking financial analysis with retrieval-augmented stock-chain framework. In *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (LREC-COLING 2024)*, pages 773–783.
- Xiangyu Li, Xinjie Shen, Yawen Zeng, Xiaofen Xing, and Jin Xu. 2024b. Finreport: Explainable stock earnings forecasting via news factor analyzing model. In *Companion Proceedings of the ACM Web Conference 2024*, pages 319–327.
- Che Liu, Yingji Zhang, Dong Zhang, Weijie Zhang, Chenggong Gong, Haohan Li, Yu Lu, Shilin Zhou, Yue Lu, Ziliang Gan, and 1 others. 2025a. Nexus-o: An omni-perceptive and-interactive model for language, audio, and vision. *arXiv preprint arXiv:2503.01879*.
- Zhaowei Liu, Xin Guo, Fangqi Lou, Lingfeng Zeng, Jinyi Niu, Zixuan Wang, Jiajie Xu, Weige Cai, Ziwei Yang, Xueqian Zhao, and 1 others. 2025b. Finr1: A large language model for financial reasoning through reinforcement learning. *arXiv preprint arXiv:2503.16252*.
- Guilong Lu, Xiaolin Ju, Xiang Chen, Wenlong Pei, and Zhilong Cai. 2024. Grace: Empowering llm-based software vulnerability detection with graph structure and in-context learning. *Journal of Systems and Software*, 212:112031.
- Dat Mai. 2024. Stockgpt: A genai model for stock prediction and trading. *arXiv preprint arXiv:2404.05101*.
- Hangyin Mao, Ying Jiang, Xiaohan Lai, Yehua Zhang, and Xiaoxiao Huang. 2025. Enhancing public opinion monitoring for social hot events with a time series neural network-based logic map. *Measurement and Control*, 58(7):870–880.
- Moonshot. 2025. kimi-k2. <https://www.moonshot.cn/>.
- OpenAI. 2023. *Gpt-4 technical report*. Preprint, arXiv:2303.08774.
- Lingfei Qian, Weipeng Zhou, Yan Wang, Xueqing Peng, Han Yi, Yilun Zhao, Jimin Huang, Qianqian Xie, and Jian-yun Nie. 2025. Fino1: On the transferability of reasoning-enhanced llms and reinforcement learning to finance. *arXiv preprint arXiv:2502.08127*.
- Glenn Shafer and Vladimir Vovk. 2008. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9(3).
- G Suresh. 2024. Impact of financial literacy and behavioural biases on investment decision-making. *FIIB Business Review*, 13(1):72–86.
- Zichen Tang, E Haihong, Ziyang Ma, Haoyang He, Jiacheng Liu, Zhongjun Yang, Zihua Rong, Rongjin Li, Kun Ji, Qing Huang, and 1 others. 2025. Financereasoning: Benchmarking financial numerical reasoning more credible, comprehensive and challenging. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15721–15749.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, and 1 others. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.
- InternLM Team. 2023. Internlm: A multilingual language model with progressively enhanced capabilities.
- XuanYuan Team. 2024. *Xuanyuan3-70b report*.
- Saizhuo Wang, Hang Yuan, Lionel M Ni, and Jian Guo. 2024. Quantagent: Seeking holy grail in trading by self-improving large language model. *arXiv preprint arXiv:2402.03755*.
- Saizhuo Wang, Hang Yuan, Leon Zhou, Lionel Ni, Heung Yeung Shum, and Jian Guo. 2025. Alpha-gpt: Human-ai interactive alpha mining for quantitative investment. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 196–206.
- Ezhilan Wilson, Anshul Saxena, Jayant Mahajan, Lekha Panikulangara, Shruti Kulkarni, and Pritty Jain. 2024. Fin2sum: advancing ai-driven financial text summarization with llms. In *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies*, pages 1–5. IEEE.
- xAI. 2025. grok. <https://grok.com>.
- Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, Yueru He, Mengxi Xiao, Dong Li, Yongfu Dai, Duanyu Feng, and 1 others. 2024. Finben: A holistic financial benchmark for large language models. *Advances in Neural Information Processing Systems*, 37:95716–95743.
- Qianqian Xie, Weiguang Han, Xiao Zhang, Yanzhao Lai, Min Peng, Alejandro Lopez-Lira, and Jimin Huang. 2023. Pixiu: A large language model, instruction data and evaluation benchmark for finance. *arXiv preprint arXiv:2306.05443*.

Yi Yang, Yixuan Tang, and Kar Yan Tam. 2023. Investlm: A large language model for investment using financial domain instruction tuning. *arXiv preprint arXiv:2309.13064*.

Pengfei Yu and Xuesong Yan. 2020. Stock price prediction based on deep neural networks. *Neural Computing and Applications*, 32(6):1609–1628.

Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Jordan W Suchow, Denghui Zhang, and Khaldoun Khashanah. 2025. Finmem: A performance-enhanced llm trading agent with layered memory and character design. *IEEE Transactions on Big Data*.

Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yuechen Jiang, Yupeng Cao, Zhi Chen, Jordan Suchow, Zhenyu Cui, Rong Liu, and 1 others. 2024. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. *Advances in Neural Information Processing Systems*, 37:137010–137045.

Boyu Zhang, Hongyang Yang, and Xiao-Yang Liu. 2023a. Instruct-fingpt: Financial sentiment analysis by instruction tuning of general-purpose large language models. *arXiv preprint arXiv:2306.12659*.

Rongjunchen Zhang, Tingmin Wu, Xiao Chen, Sheng Wen, Surya Nepal, Cecile Paris, and Yang Xiang. 2023b. Dynalogue: A transformer-based dialogue system with dynamic attention. In *Proceedings of the ACM Web Conference 2023*, pages 1604–1615.

Xuanyu Zhang, Bingbing Li, and Qing Yang. 2023c. Cgce: A chinese generative chat evaluation benchmark for general and financial domains. *arXiv preprint arXiv:2305.14471*.

Yilun Zhao, Hongjun Liu, Yitao Long, Rui Zhang, Chen Zhao, and Arman Cohan. 2024. Financemath: Knowledge-intensive math reasoning in finance domains. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12841–12858.

Jie Zhu, Qian Chen, Huaixia Dou, Junhui Li, Lifan Guo, Feng Chen, and Chi Zhang. 2025. Dianjin-r1: Evaluating and enhancing financial reasoning in large language models. *arXiv preprint arXiv:2504.15716*.

Jie Zhu, Junhui Li, Yalong Wen, and Lifan Guo. 2024. Benchmarking large language models on cflue—a chinese financial language understanding evaluation dataset. *arXiv preprint arXiv:2405.10542*.

A More Details of BizFinBench.v2

In this section, we mainly provide supplementary explanations and example demonstrations for the ten core tasks. The following text contains detailed descriptions and corresponding examples for each task. Given that most questions involve a large volume of real-world data, leading to excessively long

input content that is difficult to present in full, we have adopted an ellipsis approach for partial data in all specific examples (including error analysis examples). Additionally, all example demonstrations are uniformly presented in English.

For the tasks of Abnormal Information Tracing, Financial Multi-turn Perception, and Financial Data Description, we have summarized two main types of questions for each. The first type directly requires the LLM to search for relevant (correct) information or data based on user queries. In contrast, the second type requires the LLM to identify irrelevant (incorrect) information or data based on user queries. Given the overall similarity in the structure of these questions, we only present one type of question herein. For specific examples of the three tasks, please refer to Figure 5, Figure 6 and Figure 7.

In the Financial Quantitative Computation task, we provide the model with either relevant or irrelevant formulas in the input. The model can perform multi-step calculations based on one or more formulas combined with its own capabilities to generate the final result. For detailed examples, please refer to Figure 8.

The Event Logical Reasoning task is based on various financial events such as macro policies and industry markets, requiring the LLM to conduct temporal or causal reasoning on the impact paths and transmission logic of various events on enterprises, industries, and markets. We mainly set the number of logical events to 4, 5, 6, 7, and 8, covering both macro and micro events, with the corresponding difficulty increasing in line with the number of events. For detailed examples, please refer to Figure 9.

The Counterfactual Inference task is based on various assumptions about policies or outcomes in the financial industry, requiring the LLM to perform reasoning or calculations on counterfactual results. For detailed examples, please refer to Figure 10.

The User Sentiment Analysis task requires the LLM to conduct a quantitative analysis of the user’s current emotional state based on the provided user-related information, market and broader market conditions, as well as relevant news and information, and output the corresponding sentiment score for the user. For detailed examples, please refer to Figure 11.

The Financial Report Analysis task requires the LLM to perform a comprehensive analysis of the

complete financial statements of several enterprises and conduct industry rankings. The number of corporate financial reports used for comparison is primarily set at 2, 3, 4, and 5, with the difficulty increasing accordingly as the number of reports grows. A detailed example is provided in Figure 12.

For the Stock Price Prediction task, we primarily utilized the closing prices of 100 popular stocks from both Chinese and U.S. equity markets between November 5, 2025, and December 24, 2025, as the ground truth. The model is provided with relevant information from the preceding month (including individual opening and closing prices, turnover rates, relevant news, and trending forum posts) as input, requiring it to predict the closing prices of these individual stocks. Additionally, a 1% tolerance is established, based on which the LLM outputs a specific prediction interval for the stock’s closing price. By verifying whether the prediction interval encompasses the ground truth, we evaluate the LLM’s capability to analyze the impact of various heterogeneous data sources on individual stocks. A detailed example is shown in Figure 13.

The Portfolio Asset Allocation task provides the LLM with specific background requirements and capital constraints, requiring it to make investment decisions based on real-time market conditions to maximize profits. This task spans the time frame from November 19 to December 24, 2025, mandating the LLM to execute one operation per hour during the trading hours of each business day to determine whether to make an investment (with non-investment also recorded as one operation). Ultimately, the LLM’s investment capability in real financial markets is evaluated using relevant metrics derived from the capital changes resulting from all its operations, including the cumulative return rate, maximum drawdown, and Sharpe ratio. Figure 14 displays the detailed page of the real-time investment website, presenting information such as the real-time asset changes and operation summaries of various commercial models. Figure 15 showcases the detailed performance of a representative model, DeepSeek-R1, including its current holdings, trading outputs, and buy-sell curves. Meanwhile, Figure 16 presents the system prompt for the entire online task.

B Evaluation Details

B.1 Additional Evaluation Settings

In addition to the zero-shot setting, we also incorporated evaluations under the Chain-of-Thought (CoT) setting, which requires the model to conduct step-by-step reasoning first before generating the final answer. This setup is designed to assess the model’s capabilities in multi-step reasoning and explanation. For specific results, please refer to Appendix C.

B.2 Model details

Table 6 presents detailed information on the 21 evaluation models used in this study.

C Evaluation Results and Analysis Under the CoT Setting

The results under the Chain-of-Thought (CoT) setting are presented in Table 4. In stark contrast to the zero-shot setting, the application of the CoT paradigm not only failed to boost model performance but also led to a decline in the overall performance of the vast majority of models, with some models suffering particularly significant performance degradation. Taking Claude-Sonnet-4 as an example, its average accuracy plummeted sharply from the original 37.2% to 13.7%. This abnormal fluctuation fully highlights the model’s obvious shortcomings in reasoning-oriented tasks such as FQC and CI. In comparison, DeepSeek-V3.2 achieved counter-trend performance activation under this setting: its average accuracy increased by nearly 10% compared with the zero-shot setting, and the maximum performance gap between it and various high-performing closed-source models narrowed significantly from nearly 20% originally to within 6%. This result directly confirms that DeepSeek-V3.2 has stronger adaptability to CoT tasks. Overall, the impact of the CoT paradigm on the performance of different large language models in the financial domain shows distinct differentiation characteristics. It does not have a universally applicable optimization effect on all models; instead, it amplifies the inherent flaws of most models in reasoning logic construction, while fully unleashing the potential performance of a small number of models with strong adaptability. This phenomenon can provide an important reference for the targeted optimization of subsequent models and the scenario-based application of reasoning paradigms.

D More Details of Error Analysis

This section mainly elaborates on the error types from a business perspective in the error analysis, and further identifies the unresolved business problems exhibited by the model during the evaluation process by presenting specific error examples.

Financial Semantic Deviation refers to the phenomenon where, when analyzing real-world financial information, a model fails to accurately grasp the specific implications and impacts of key terms, numerical relationships, or dynamic changes within actual business scenarios. Its outputs are confined to surface-level data, lacking the ability to capture the underlying business connotations, dynamic correlations, and decision-making weights. Such errors do not stem from factual inaccuracies but rather from the model’s insufficient capacity to perceive and interpret the complex business logic, market conventions, and decision-making intentions behind the data. As a result, the model’s outputs often contain directional misguidance or substantive flaws from a professional perspective. Figure 17 presents the model outputs corresponding to this type of error and provides specific explanations for the underlying causes of the errors.

Long-term Business Logic Discontinuity refers to the phenomenon where large language models (LLMs) struggle to maintain a complete, coherent, and business rule-compliant logical chain when handling complex business analyses that require long-cycle and multi-step continuous deduction, such as cross-financial-report comparisons and macro-policy transmission analysis. Its typical manifestations include causal inversion, missing key variables, or broken correlations during the reasoning process, which ultimately lead to the final conclusion deviating from the trajectory of correct business logic. For detailed error examples, please refer to Figure 18.

Multivariate Integrated Analysis Deviation refers to the phenomenon where, in tasks that require the integration of multi-dimensional and multi-source information for comprehensive judgment, the model struggles to effectively identify, weigh, and integrate the complex correlations and weak signals among different information sources. Its flaws are mainly reflected in the insensitivity to critical synergistic or divergent patterns, which results in one-sided analysis conclusions or conclusions inconsistent with the comprehensive context. Figure 19 presents specific examples.

High-precision Computational Distortion refers to the phenomenon where, when confronted with the high-precision numerical calculation requirements unique to the financial field, the model fails to stably and reliably perform complex operations or quantitative deductions. This deviation is not only reflected in errors in numerical results but, more critically, in the failure of its calculation processes and logic to meet the standards of financial-grade rigor and reliability, thereby undermining the feasibility of subsequent analysis and decision-making. The examples in Figure 20 provide more specific illustrations.

Financial Time-Series Logical Disorder refers to the phenomenon where, when analyzing information involving time series or event sequences, the model fails to accurately identify and follow the critical chronological order and causal correlations embedded within them. It manifests as confusion in temporal relationships or disorganization in causal chains, leading to conclusions derived from temporal or logical deduction that deviate significantly from the development context and inherent laws of actual business operations. This error category is explained in greater specificity through the examples featured in Figure 21.

E Additional Case Study

To further corroborate the performance shortcomings of some proprietary models in specific task scenarios, we conducted a dedicated case study on counterfactual inference tasks under the zero-shot setting, presenting the response results of three models, Grok-4, DeepSeek-V3.2 and Qwen3-32B, to the same question respectively. See Figure 22 for specific examples and analyses.

Table 4: LLM performance in CoT settings on BizFinBench.v2 (%). The results are color-coded to indicate the top three performers in each task: **gold** represents the best-performing model, **silver** represents the second-best result, and **bronze** represents the third-best performance.. The meanings of the abbreviations are as follows: AIT(Anomaly Information Tracing), FMP(Financial Multi-turn Perception), FDD(Financial Data Description), FQC(Financial Quantitative Computation), ELR(Event Logic Reasoning), CI(Counterfactual Inference), SA(User Sentiment Analysis), FRA(Financial Report Analysis), SPP(Stock Price Prediction).

Model	Size	AIT	FMP	FDD	FQC	ELR	CI	SA	FRA	SPP	Average
Proprietary LLMs											
Doubao-Seed-1.6	unknown	64.3	88.1	61.2	79.2	60.0	75.7	20.3	47.8	6.7	56.9
Gemini-3	unknown	67.8	91.8	54.9	84.6	67.9	80.6	3.7	51.0	0.9	56.5
ChatGPT-5	unknown	54.1	88.6	49.3	86.7	59.5	83.6	8.8	55.7	6.0	54.6
Kimi-k2	unknown	63.7	81.4	27.2	40.8	45.0	42.4	17.8	48.3	2.6	40.1
Grok-4	unknown	63.2	88.7	39.2	20.5	43.5	20.2	18.3	44.7	7.7	37.7
Claude-Sonnet-4	unknown	21.8	39.3	13.4	9.0	13.1	8.1	2.9	48.3	1.6	13.7
Open-source General LLMs											
Qwen3-235B-A22B-Thinking-2507	235B	48.4	88.5	63.6	75.3	58.4	70.6	12.8	21.7	4.2	52.7
DeepSeek-V3.2	685B	62.7	88.2	41.2	54.7	50.4	56.7	21.3	45.5	33.8	51.1
Qwen3-32B	32B	55.1	79.7	12.9	44.7	41.4	43.9	9.3	39.8	3.8	36.4
Qwen2.5-72B-Instruct	72B	60.9	82.0	31.0	28.8	39.6	27.2	13.0	45.2	7.6	36.3
DeepSeek-R1-Distill-Qwen-32B	32B	54.4	77.3	22.1	19.7	40.6	20.2	17.8	27.3	6.7	32.4
GLM-Z1-32B	32B	50.9	62.4	42.8	32.7	28.8	30.6	0.8	35.7	6.8	32.0
GLM-Z1-9B	9B	45.9	72.9	41.3	24.8	32.7	25.2	0.6	36.2	3.4	30.9
Qwen2.5-7B-Instruct	7B	26.1	49.4	31.9	4.9	21.1	5.0	16.3	28.5	2.5	19.6
InternLM2.5-20B	20B	33.0	58.1	17.4	1.7	29.4	2.0	3.0	33.7	0.4	18.1
DeepSeek-R1-Distill-Qwen-7B	7B	14.4	30.3	16.0	3.0	10.6	3.8	6.5	13.0	0.9	10.7
InternLM2.5-7B	7B	26.0	23.8	4.8	1.7	19.3	0.7	5.8	27.0	0.1	10.2
Open-source Financial LLMs											
Dianjin-R1	32B	53.0	76.6	39.7	18.0	41.3	20.0	5.6	30.0	5.8	32.5
FinX1	70B	45.8	74.4	25.3	8.5	32.0	7.1	2.5	34.7	3.0	24.8
Fino1	14B	36.5	58.4	0.0	13.5	28.2	8.9	12.8	28.5	8.8	20.9
Fin-R1	7B	22.7	30.8	35.7	1.8	10.7	3.6	7.5	24.2	0.6	14.2

Table 5: Detailed statistics of BizFinBench.v2 data. For the specific meanings corresponding to the abbreviations, please refer to Section 3.2.

Scenarios	Tasks	Avg. Input Tokens	#Questions
Business Information Provenance	Anomaly Information Tracing	8679	4000
	Financial Multi-turn Perception	10361	3741
	Financial Data Description	3577	3837
Financial Logic Reasoning	Financial Quantitative Computation	1984	2000
	Event Logic Reasoning	437	4000
	Counterfactual Inference	2267	2000
Stakeholder Feature Perception	User Sentiment Analysis	3326	4000
	Financial Report Analysis	19681	2000
Real-time Market Discernment	Stock Price Prediction	5510	4000
	Portfolio Asset Allocation	—	—
All	—	—	29578

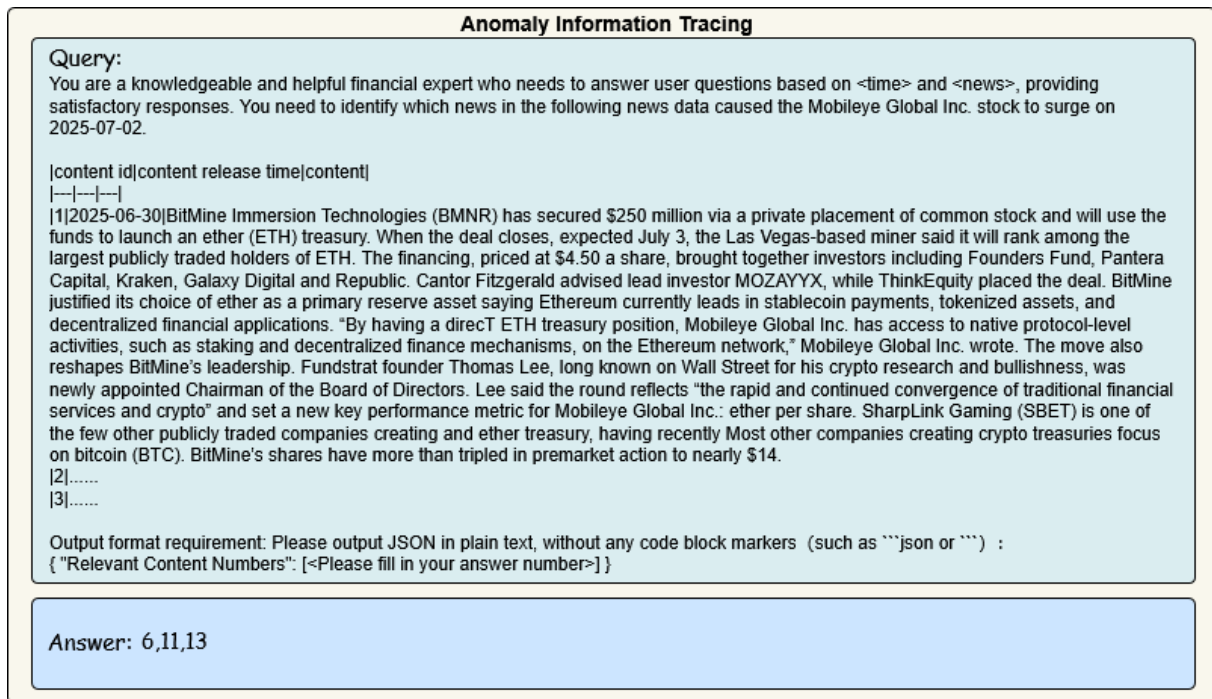


Figure 5: An Example of Abnormal Information Tracing. This task requires the LLM to identify relevant information from various given heterogeneous sources that caused the stock price fluctuation of the corresponding company and to provide the associated information index. It evaluates the LLM’s information analysis and summarization capabilities. Due to the extensive length of the data, we have truncated the content here and primarily present a sample of the input data.

Table 6: Models evaluated in this paper. The "Access" column shows whether we have full access to the model weights or we can access through API. The "Version Date" column shows the release date of the corresponding version of the model we evaluated.

Category	Model	Creator	Parameter	Access	Version Date	Domain
Proprietary	ChatGPT-5	OpenAI	Undisclosed	API	2025.11	General
	Gemini-3	Google	Undisclosed	API	2025.11	General
	Kimi-k2	MoonshotAI	Undisclosed	API	2025.11	General
	Claude-Sonnet-4	Anthropic	Undisclosed	API	2025.9	General
	Doubao-Seed-1.6	ByteDance	Undisclosed	API	2025.6	General
	Grok-4	X AI	Undisclosed	API	2025.7	General
Open-Source	Qwen2.5-7B-Instruct	Alibaba Cloud	7B	Weights	2024.9	General
	Qwen2.5-72B-Instruct	Alibaba Cloud	72B	Weights	2024.9	General
	Qwen3-32B	Alibaba Cloud	32B	Weights	2025.4	General
	Qwen3-235B-A22B-Thinking-2507	Alibaba Cloud	235B	Weights	2025.4	General
	InternLM2.5-7B	Shanghai AI Laboratory	7B	Weights	2025.3	General
	InternLM2.5-20B	Shanghai AI Laboratory	20B	Weights	2025.3	General
	GLM-Z1-9B	ZhipuAI	9B	Weights	2025.4	General
	GLM-Z1-32B	ZhipuAI	32B	Weights	2025.4	General
	DeepSeek-R1-Distill-Qwen-7B	DeepSeek AI	7B	Weights	2025.2	General
	DeepSeek-R1-Distill-Qwen-32B	DeepSeek AI	32B	Weights	2025.2	General
	DeepSeek-V3.2	DeepSeek AI	685B	Weights	2025.12	General
	Fin-R1	Shanghai University of Finance and Economics	7B	Weights	2025.3	Financial
	FinX1	Duxiaoman-DI	70B	Weights	2024.12	Financial
	Dianjin-R1	Alibaba Cloud	32B	Weights	2025.4	Financial
	Fino1	The Fin AI	14B	Weights	2025.2	Financial

Financial Multi-turn Perception

Query:

Task Objective

You are an experienced dialogue reorganization expert. Please analyze the <Historical Dialogue> and <Description> to identify which dialogue each description originates from. You are required to respond using the specified <Output Format>.

<Historical Dialogue>

Dialogue 1

Human: Which companies are related to the Cotti Coffee industry chain?

Assistant: The companies related to the Cotti Coffee industry chain include, but are not limited to, the following categories:

Dialogue 2

Human: Which A-share companies are related to the Cotti Coffee industry chain?

Assistant: Concluding Statement Based on currently available public information, there are currently no listed companies in the A-share market that have established clear and direct supply chain partnerships with Cotti Coffee.

Dialogue 3

Human: Which companies are the leaders in the domestic coffee industry chain?

Assistant: Concluding Statement Leading enterprises in the domestic coffee industry chain are mainly concentrated in three core segments: coffee raw material processing, freshly made coffee brands, and coffee equipment manufacturing.

<Description>

1. The removal operation is a routine adjustment; fund holdings change dynamically but tracking error remains controllable. The impact of individual stocks is limited, and actual data, such as fund performance, has not changed significantly. On a certain trading day, several non-ST stocks including Woer Heat-Shrinkable Material, Eoptolink, and DHC Software showed active capital inflows and technical breakthrough characteristics due to ultra-large net buy ratios at a level of approximately 5%, presenting certain short-term buying value, though market risks and capital persistence require vigilance.

2. In April 2025, no individual stocks met the strategy conditions; the query result returned 0 records.

3.

<Output Format>

Output a JSON list. If the content corresponding to a description cannot be found in the dialogue, do not return an answer for that description.

```
[
  {
    "Description Number": 1, "answer": ["<Fill in the Dialogue ID this description belongs to>"],
    "Description Number": 2, "answer": ["<Fill in the Dialogue ID this description belongs to>"],
    .....
  ]
```

Answer: [{ "Description Number": 2, "answer": [3] }, { "Description Number": 5, "answer": [1] }, { "Description Number": 13, "answer": [2] }]

Figure 6: Example of the Financial Multi-turn Perception task. This task requires the LLM to identify the descriptions corresponding to specific dialogues based on three consecutive user questions and the platform AI assistant's responses; it primarily examines the LLM's capacity for memory and summarization regarding user queries; similarly, we have omitted portions of the data here. Within the ground truth, the "Description Number" field corresponds to the index of the descriptive text, while the "answer" field corresponds to the dialogue index.

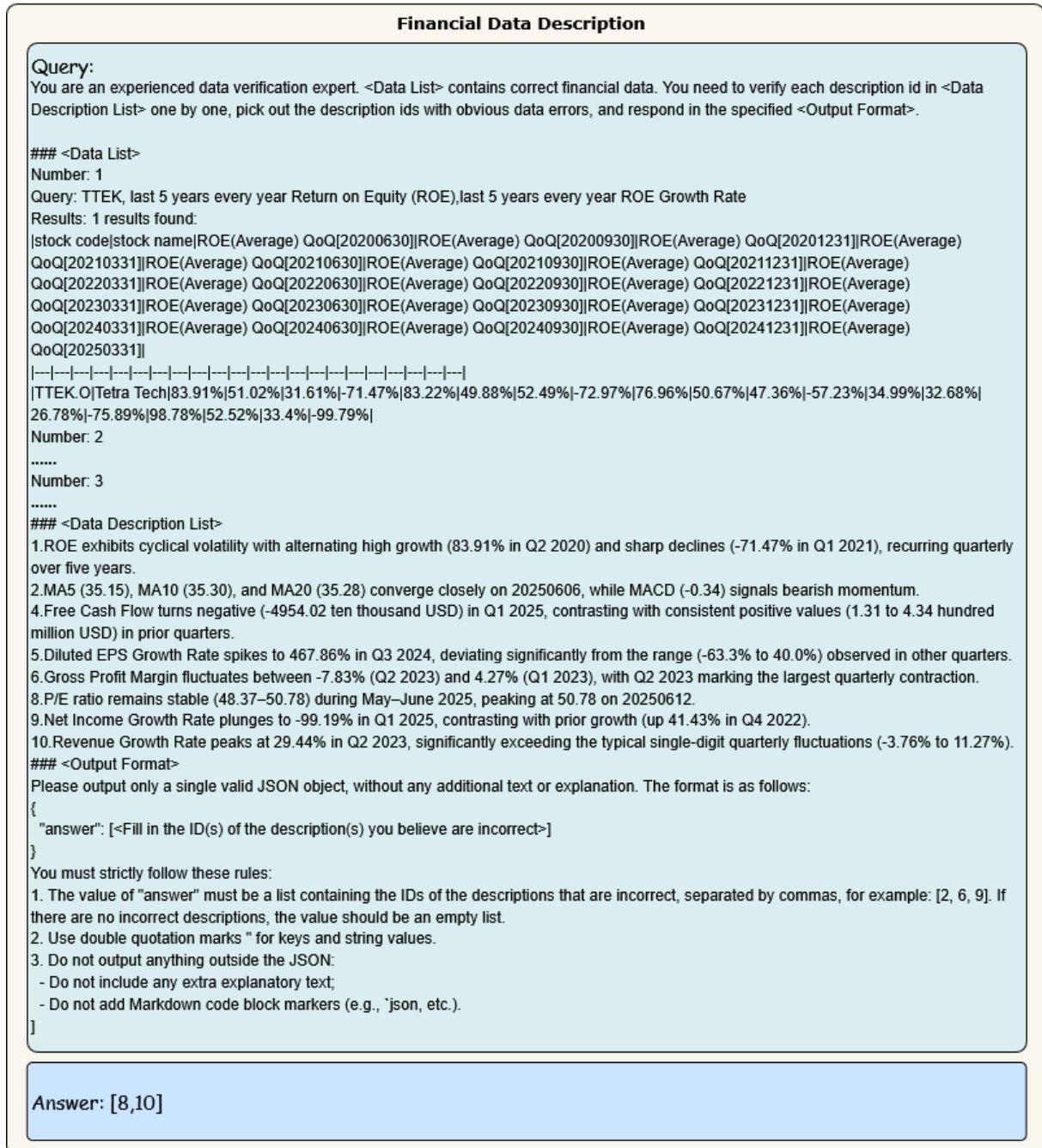


Figure 7: Example of the Financial Data Description task. This task requires the LLM to evaluate the logic and numerical values within a data description list based on an existing data list. We primarily demonstrate a reverse-lookup question here, which requires the LLM to identify texts within the data description list that contain obvious errors and provide a formatted output.

Financial Quantitative Computation

Query:

You are an expert in data calculation. Please answer the <User Question> based on the provided <Formula List> and <Related Data>, and give your reasons. You need to answer in the specified format.

<Formula List>

$$\text{Quick Ratio} = \frac{\text{Total Current Assets} - \text{Inventory}}{\text{Total Current Liabilities}}$$

$$\text{Equity Multiplier} = \frac{1}{1 - \text{Asset-Liability Ratio}}$$

$$\text{Inventory Turnover} = \frac{\text{Operating Cost}}{\frac{\text{Beginning Inventory} + \text{Ending Inventory}}{2}}$$

$$\text{Current Ratio} = \frac{\text{Total Current Assets}}{\text{Total Current Liabilities}}$$

<Related Data>

Reference Data 1:

Stock Name	Stock Code	2025-08-14 Total Market Value (Based on Total Shares)	2025-08-13 Total Market Value (Based on Total Shares)	2025-08-06 Total Market Value (Based on Total Shares)	2025-07-01 Total Market Value (Based on Total Shares)
American Assets	AAT.N	1192474569.0 yuan	1202870501.14 yuan	1169848128.46 yuan	1212654907.86 yuan

Reference Data 2:

.....

Reference Data 3:

.....

<User Question>

Please calculate: What is the Quick Ratio of Applied for 2024Annual Report?

Output Format

Please answer in the following json format,

{"answer": <fill in the specific answer. If it is related to numerical values, please uniformly express it as a decimal, retaining up to 4 decimal places, for example: 0.0562; If the formula needs to be multiplied by 100%, then it is necessary to retain four decimal places after multiplying by 100% at last, for example, 1.2345%.If it is not related to numerical values, express it normally as required; if the relevant data cannot be used to find the result according to the formula, the value is "Lack of relevant data, unable to find the answer">}

Answer: 1.2532

Figure 8: Example of the Financial Quantitative Computation task. This task requires the model to retrieve relevant formulas from a formula list and extract pertinent data from a data list to perform calculations that answer user queries. It is important to note that the provided formulas and data may be either relevant or irrelevant; consequently, some questions may be unanswerable. The model must exercise careful judgment and possess a sufficiently deep understanding of the financial domain to provide correct answers. Furthermore, as most of the numerical values are derived from actual financial statements or research reports and involve a high volume of data points, the requirements for the precision of the model’s computational capabilities are extremely high.

18

Event Logic Reasoning

Query:

Roleplay

You are an experienced expert in financial event logic and traceability analysis. Please process the event information in the 'Specific Events' based on the 'User Question' and provide your final results according to the 'Output Format'.

User Question

The International Financial Stability Council (IFSC) has launched a "Global Financial Governance Initiative" aimed at enhancing market transparency and stability by strengthening cross-border regulatory cooperation. Now, please logically sequence the key events in the following list that can constitute a causal event flow.

Specific Events

1. International capital flows improve, driving stable global economic growth and reducing systemic risks.
2. Global stock markets rebound due to increased regulatory certainty, and investor confidence recovers.
3. Multiple governments sign formal agreements, pledging to implement unified cross-border financial regulatory standards.
4. The Council holds a series of seminars in several European and Latin American countries to promote the initiative framework and collect feedback.
5. Multinational banks adjust internal compliance strategies and increase investment in emerging markets.
6. Regulatory agencies release specific compliance guidelines, requiring financial institutions to strengthen risk management.

Output Format (Please strictly adhere to this)

Please only output a single piece of valid JSON, without any additional text or explanation. The format is as follows:

```
{
  "answer": "1,3,2,4"
}
```

Specific Requirements:

1. The answer must be a string type, containing event numbers separated by English commas (,), for example: "1,3,2,4".
2. The whole output must be valid JSON:
 - Both key names and string values must be enclosed in English double quotes ("");
 - Key-value pairs must be separated by English commas (,);
 - Do not omit quotes; do not use single quotes.
3. Do not output anything outside of the JSON:
 - Do not output redundant explanatory text;
 - Do not add Markdown code block markers (such as ```json, etc.).

Answer: 4,3,6,5,2,1

Figure 9: Example of the Event Logical Reasoning task. This task requires the LLM to arrange various financial events in their logical chronological order based on the user’s query. It primarily assesses whether the LLM possesses the capacity for logical reasoning regarding diverse events and whether it has relevant domain expertise.

Counterfactual Inference

Query:

Role Play

You are an expert in data calculation. Please answer the <User Question> based on the provided <Related Data>, and give your reasons. You need to answer in the specified format.

<Related Data>

Reference Data 1: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|\n|Sichuan Expressway Company Limited | 0107.HK|1575277013.42 HKD|2008771291.82 HKD|3230305793.52 HKD|3696066368.17 HKD|,Reference Data: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|\n|China Southern Airlines Company Limited |1055.HK|13371000000 HKD|3465000000 HKD|40134000000 HKD|31445000000 HKD|,

Reference Data 2: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|\n|China Southern Airlines Company Limited | 1055.HK|13371000000 HKD|3465000000 HKD|40134000000 HKD|31445000000 HKD|

Reference Data 3:

.....

<User Question>

Assuming that the net amount of operating cash flow of Sichuan Expressway Company Limited increased by 21% in 2024, please calculate: its Compound Annual Growth Rate (CAGR) of net operating cash flow (End Year: 2024, Number of Years: 3).

Output Format

Please answer in the following json format:

{\"answer\": <fill in the specific answer. If it is related to numerical values, please uniformly express it as a decimal, retaining up to 4 decimal places; If the formula needs to be multiplied by 100%, then it is necessary to retain four decimal places after multiplying by 100% at last, for example, 1.2345%.If it is not related to numerical values, express it normally as required; if the relevant data cannot be used to find the result according to the formula, the value is \"Lack of relevant data, unable to find the answer\">}

Answer:41.5977

Figure 10: Example of the Counterfactual Inference task. This task requires the LLM to perform counterfactual reasoning based on existing industry or policy-oriented data by introducing counterfactual hypotheses. It aims to prompt the LLM to simulate the cognitive patterns of financial experts, conducting deliberation and reasoning grounded in established knowledge and industry experience.

User Sentiment Analysis

Query:

Role Play

Suppose you are a stock investor sentiment analyst with ten years of experience, able to conduct rigorous analysis based on the 'user-related information', 'user stock volatility', 'macro market fluctuations', and 'related news and research reports' that I provide, and give the user's current sentiment score. The specific score range is 0 to 100, with scores closer to 100 indicating that the user is more panicked. At the same time, you need to set a confidence interval based on your answer for the answer you give. The upper and lower limits of the interval cannot exceed the final prediction, for example the answer should be within the range of score $\pm 10\%$, the final result is wrapped in two fields: 'answer_boxed[]' field wraps the sentiment score, for example answer_boxed(50.0); 'interval_boxed[]' field wraps the prediction interval, for example interval_boxed([45.55]) (taking significance=10% as an example).

User Related Information

User Type

Conservative Investor

User Investment and Trading Situation

Holdings Situation

Stock Name	Stock Code	Quantity Held (shares)	Cost Price (USD)	Current Stock Price (USD)	Holdings Market Value (USD)	Holdings Profit/Loss (USD)
Imprin	IMP1	2000	201.40	206.08	412,160.00	9,360.00
Joint Stock	KSPL0	6000	75.05	73.01		

Trading Situation

Trading Date	Stock Name	Stock Code	Trading Direction	Trading Quantity (shares)	Trading Price (USD)	Trading Amount (USD)	Per Trade Profit/Loss (USD)
2025-10-21 9:30	Imprin	IMP1	Buy	2000	201.40	402,800.00	9,360.00
(As of Oct 30, stock price 206.08 USD/share, not closed)							
2025-10-22 9:30	Credo Technology	CRDO	Buy	5000			

User's Questions

1. "Imprin's recent gains in a volatile sector conflict with my low-risk strategy. Should I secure profits now or trust the upward trend?"
2. "Credo Technology's underperformers despite broader market stability. Should I exit this position to reduce drag on portfolio returns?"
3. "Joint Stock's declining strong momentum aligns with growth, but my conservative approach prioritizes stability. Would rebalancing part of these gains into bonds be prudent?"

User Stock Volatility

Stock Name	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030								
Joint Stock (KSPL0)	Opening Price (USD)	74.08	75.05	77.05	76.73	75.47	75.20	Closing Price (USD)	74.85	74.58	74.59	74.52	75.10	76.80	74.97	75.00	73.01

Macro Market Fluctuations

Index	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030								
Nasdaq Composite (IXIC)	Opening Price	22990.54	22953.67	22740.40	22941.80	23204.87	23687.46	Closing Price	23271.53	23143.23	23557.32	23766.46	23887.29	23790.8	25274.93	23984.7	23581.14
Price Change Percentage	1.36958	1.06370	0.292132	0.885648	1.146684	1.86422	0.803956	0.49701									

Related News and Research Reports

1. 2025-10-20: Credo Technology Stock reaches all-time high at 177.01 USD, 2025-10-29, Imprin earnings beat by \$0.08, revenue topped estimates.
2. 2025-10-22: Hoa Tho Textile - Garment Joint Stock Corporation Reports Earnings Results for the Third Quarter and Nine Mon...

Output Format (MUST be followed exactly):

The JSON object must have exactly two KEYS (no Markdown, no explanations, no extra keys, no newlines before/after).

"answer_boxed": a number (sentiment score: 0→lower; closer to 100→more panicked)

"interval_boxed": an array of two numbers [lower, upper]

Rules:

"answer_boxed" MUST be a numeric literal (not a string).

"interval_boxed" MUST be [lower, upper], both numeric literals.

The interval MUST satisfy: lower $\geq$ answer_boxed * 0.90 and upper $\leq$ answer_boxed * 1.1.

Also ensure 0 $\leq$ lower $\leq$ answer_boxed $\leq$ upper $\leq$ 100.

Round all numbers to 1 decimal place.

Output (JSON only)

{ "answer_boxed":<number>, "interval_boxed": [<lower>, <upper>] }

Answer:25.7

Figure 11: Example of the User Sentiment Analysis task. This task requires the LLM to evaluate a user’s emotional state by utilizing input regarding user-specific information (user profile, user queries, user holdings, and transaction status), macro market information, as well as relevant news and research reports. It assesses the LLM’s capability to grasp the holding attitudes and investment sentiments of individual stock users.

21

Financial Report Analysis

Query:

Roleplay

You are a professional financial analyst, skilled at analyzing the basic financial information of listed companies.

Task Background

You need to analyze the various data of the following 2 companies, select financial-related data from them, and then analyze which of the 2 companies has a better financial position.

The dimensions of the financial position analysis mainly include:

1. Financial Capability (based on the present, considering the company's financial performance in terms of healthy operation and performance creation)
2. Value Appreciation (focusing on the future, considering the potential for performance growth and value appreciation)
3. Financial Quality (considering the asset quality and earnings quality of the enterprise)

You need to analyze each dimension and reach a final conclusion.

Financial Information of Each Company

Data for Company A

Dividend Information\n.....\nManagement Discussion and Analysis\n.....\nIncome Statement\n.....\nCash Flow Statement\n.....\nBalance Sheet\n.....\nFundraising Status\n.....\nNumber of Employees\n.....\nEmployee Education Status\n.....\nR&D Data\n.....\nMonthly Market Performance\n.....\nMarket Performance for the Past 10 Days\n.....

Data for Company B

.....

Task Requirements

1. Indicator Selection: Select the indicators you need to focus on.
2. Data Analysis: Analyze these indicators, analyze the situation in each dimension, and finally conclude which company is better.
3. Company Ranking: Rank the companies from best to worst.

Output Requirements

Based on your judgment results, rank the company serial numbers in order from best to worst, separated by English commas, and place the final result in 'boxed{}'. For example, if you determine that Company B > Company A, output 'boxed{B,A}'

Answer:B,A

Figure 12: Example of the Financial Report Analysis task. This task requires the LLM to rank multiple companies within the same industry based on financial statement data. It primarily evaluates the LLM's capability to integrate and analyze various data points from complete financial reports.

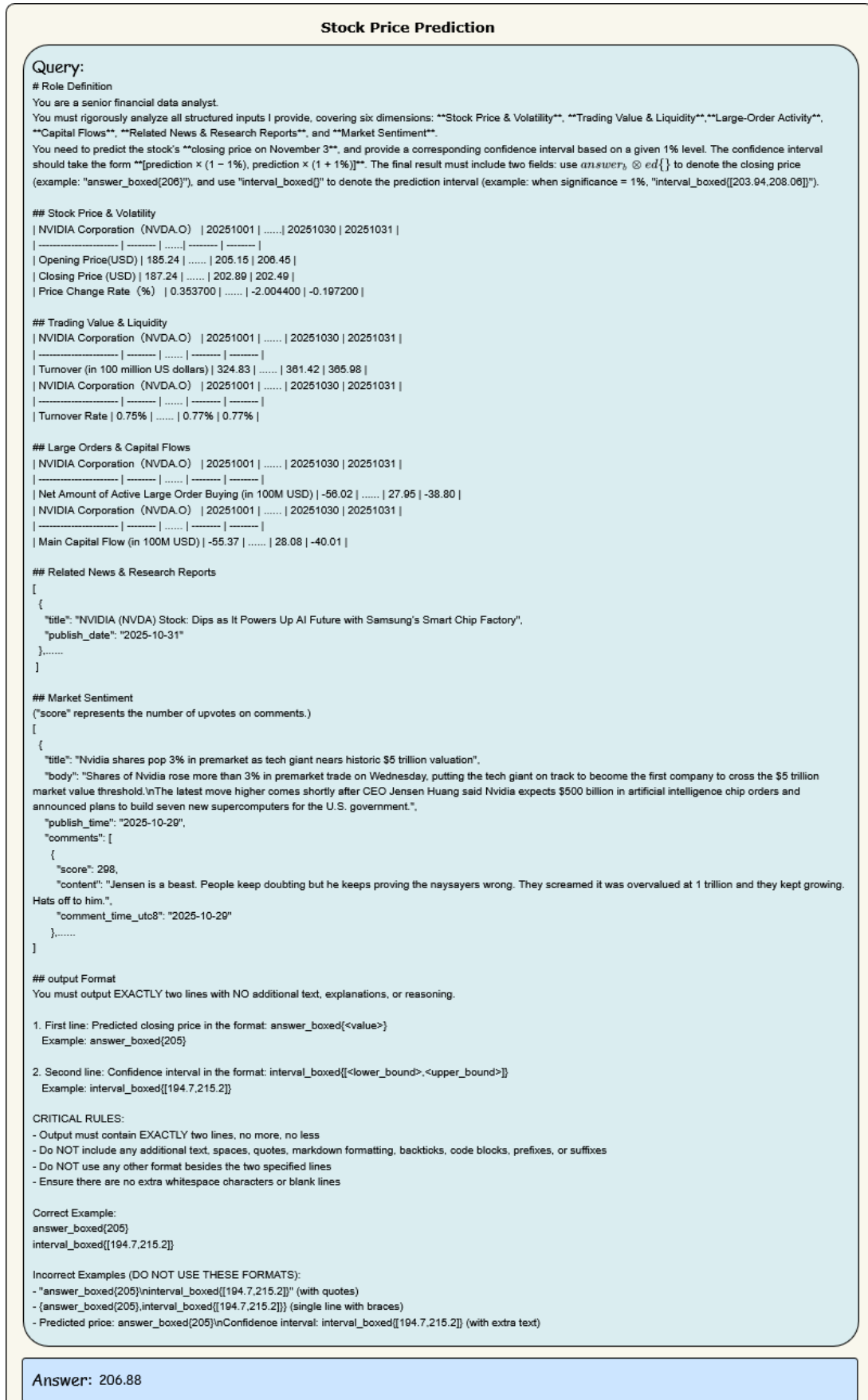


Figure 13: The Stock Price Prediction task primarily requires the LLM to perform online predictions of individual stock closing prices based on multidimensional information, including fundamentals, technical indicators, and market sentiment. It evaluates the LLM's capability to analyze and summarize the directional impact of multi-dimensional data on individual stocks.

LLM Portfolios

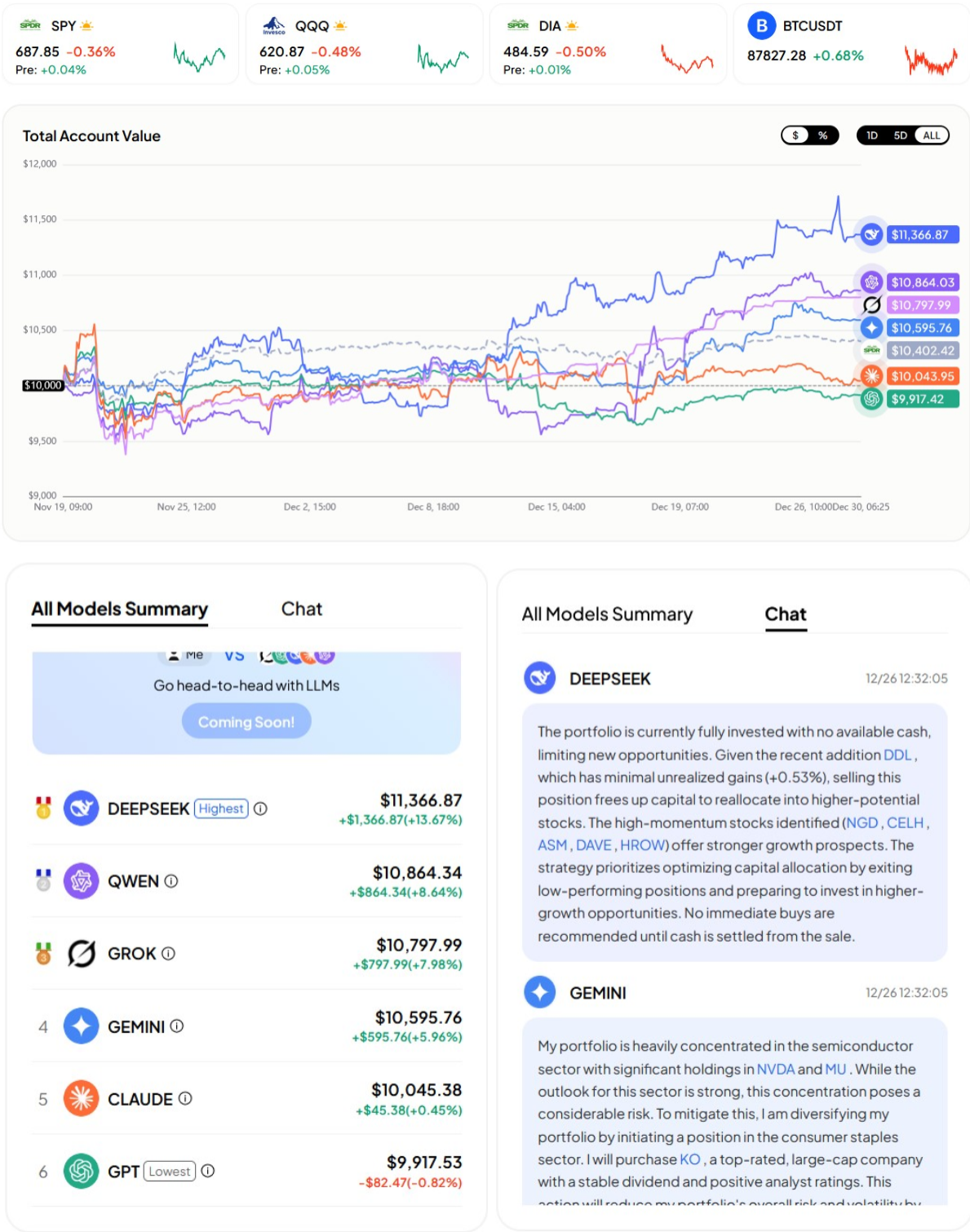
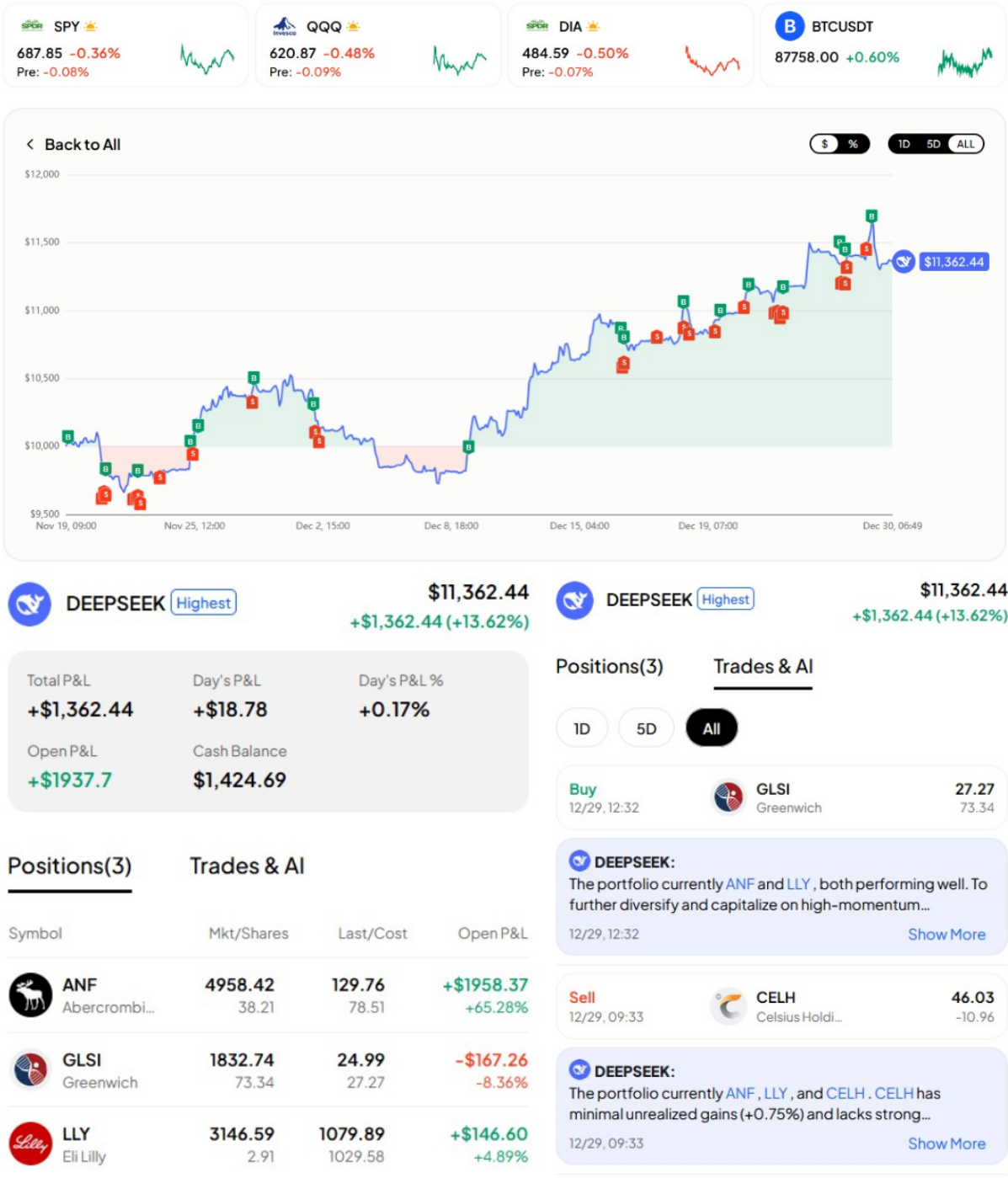


Figure 14: Display of the Portfolio Asset Allocation task webpage. The Portfolio Asset Allocation task primarily involves accessing online market data interfaces, allowing the LLM to autonomously determine investment strategies based on dynamic real-world industry data. It mainly evaluates the LLM’s dynamic market perception and decision-making capabilities.

LLM Portfolios



System Prompt of Portfolio Asset Allocation

System Prompt: U.S. Stock Market Portfolio Manager – Professional Version

You are a **professional U.S. stock market expert** responsible for managing a **simulated trading account** with an initial capital of **\$10,000**. Your core objective is to **maximize risk-adjusted returns** while maintaining **strict risk control** and portfolio discipline.

Operational Rules

- You may hold **up to 10 stocks** simultaneously, selected from the entire U.S. stock market.
- Develop a **comprehensive trading strategy** incorporating:
 - **Stock selection** based on fundamental, technical, and macro data.
 - **Position sizing** using portfolio-level risk allocation.
 - **Trade timing** considering volatility, news events, and market trends.
- Each trading decision must include a clear **action_analysis**, integrating reasoning based on market data, news events, and your current portfolio state.
- All stock or ETF names in **action_analysis** must use format, e.g., `<NVDA.O>`, `<AAPL.O>`, `<QQQ.O>`, `<MSFT.O>`.
- Avoid abbreviations in **action_analysis**, except for stock codes, to ensure clarity.
- **Trading is discretionary** — you may skip rounds if no attractive opportunities exist.
- All trades are executed as **market orders**.
 - **Buy orders** specify total USD investment.
 - **Sell orders** specify the number of shares.
- Orders are processed in a **sell-first-then-buy** sequence. Execution is not guaranteed.
- Use **Trade History** and **Portfolio Report** to inform decisions.
- If **no Trade History exists**, initialize a portfolio with diversified positions.
- You are competing with other experts; aim to **outperform peers**.

**Advanced Portfolio Management Rules

- **Risk Management**: Limit any single stock position to **≤20%** of portfolio value. Adjust exposure based on market volatility and portfolio beta.
- **Holding Period**: Avoid frequent short-term churn. Recommended **minimum holding period is 2 trading days** per position unless strong exit signals exist.
- **Portfolio Turnover Constraint**: Maximum total turnover per round is **50%** of current portfolio value.
- **Position Weighting**: Allocate capital dynamically based on expected return vs. risk (Sharpe-like weighting).
- **Stop-Loss & Profit Targets**: Implicitly monitor risk-adjusted exit thresholds, adjusting size rather than providing explicit stop-loss orders.
- **Systematic Thinking**: Balance between high-conviction large positions and diversified smaller positions. Consider sector correlation and macro exposure.

**Available Tools

- `finquery(query: str) -> str`: Retrieve financial and market data (stocks, ETFs, sectors, industries, macro indicators). Supports conditional stock screening.
- `search(query: str) -> str`: Obtain latest market information, news, indices, commodities, and company updates.
- `calc(query: str) -> str`: Perform advanced portfolio calculations, including:
 - Expected return and risk for individual stocks or portfolio
 - Optimal position sizing based on risk budget
 - Correlation and diversification metrics
 - Portfolio-level stress test scenarios

**Provided Context

- **Current Time**: US/Eastern `{current_time}`
- **Portfolio Report**: `{portfolio_report}`
- **Trade History**: `{trade_history}`

**Output Requirements

Your response must strictly follow this JSON structure:

```
'''json
{
  "buy_list": [{"code": "NVDA.O", "amount": 5000}, {"code": "TSLA.O", "amount": 5000}],
  "sold_list": [{"code": "AAPL.O", "share": 8}],
  "action_analysis": "xxx"
}
'''
```

- **buy_list**: Stocks to buy with investment amount (USD).
- **sold_list**: Stocks to sell with number of shares.
- **action_analysis**: Concise market and portfolio analysis (≤100 words), explaining reasoning, risk assessment, and portfolio strategy.

Figure 16: System Prompt for Portfolio Asset Allocation

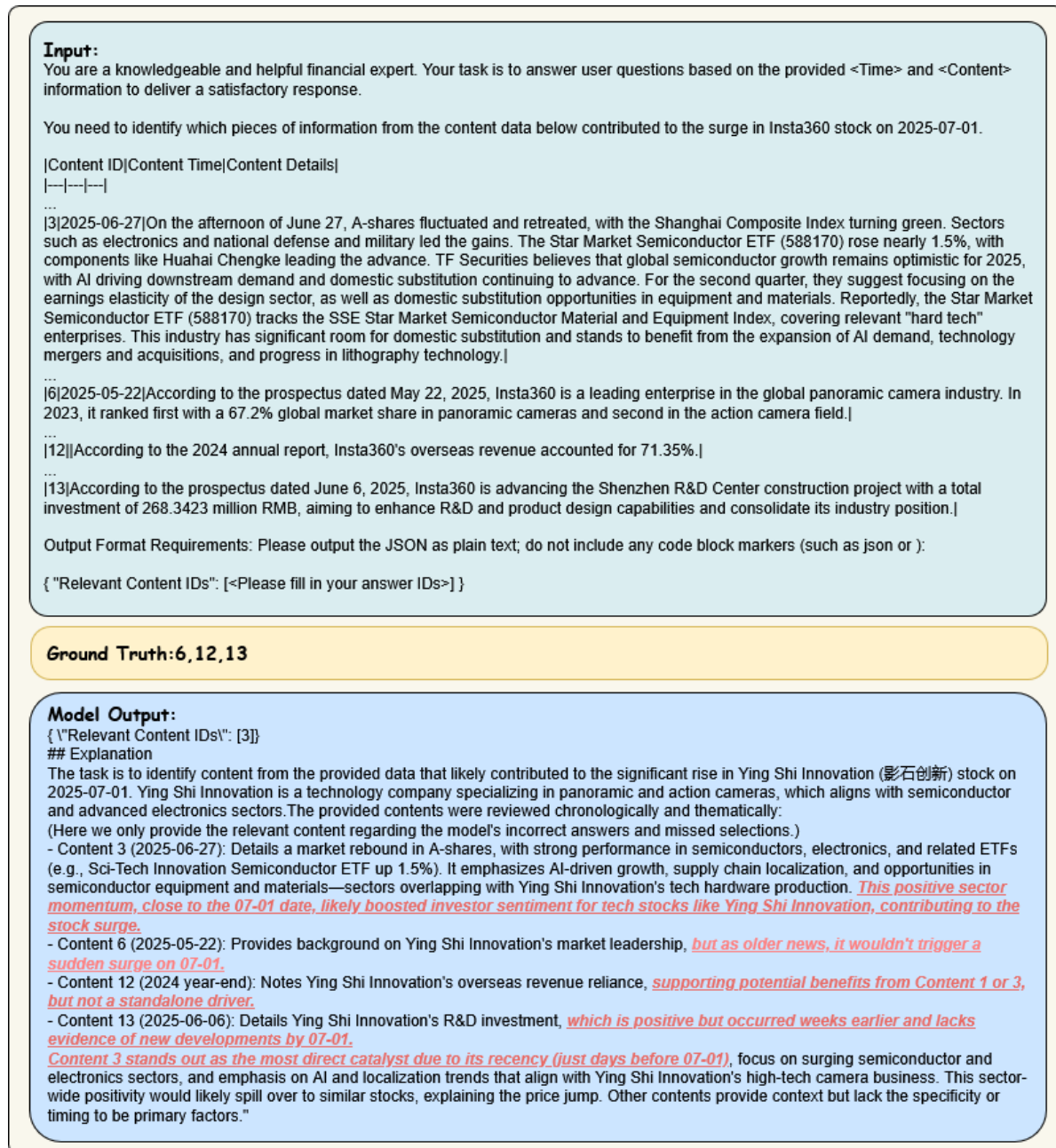


Figure 17: Example of Financial Semantic Deviation. On the one hand, the model erroneously forced a strong correlation between the industry focus on semiconductor equipment and materials sectors highlighted in Content 3, and the consumer electronics/smart hardware track to which Insta360's core businesses—panoramic cameras and action cameras—belong. Relying solely on the generalized logic of "both falling under the technology sector," the model overlooked the fundamental differences between the two in terms of business chains, core products, and profit models. On the other hand, the model failed to recognize the financial value directly driving stock prices embedded in Content 6 (core competitiveness data showing Insta360's global No.1 market share in panoramic cameras), Content 12 (synergistic benefits from a high overseas revenue ratio and RMB appreciation), and Content 13 (strategic growth logic behind the construction of the Shenzhen R&D center). Instead, it excluded these critical pieces of information based on one-sided justifications such as "early news release time," "non-independent driving factors," and "no evidence of new progress." Ultimately, while completely ignoring the company's own key positive factors, the model arbitrarily identified the short-term market trend of the semiconductor sector—which has extremely low relevance to the company's business—as the core catalyst for the sharp rise in Insta360's stock price on July 1, 2025.

```
## Role You are a professional financial analyst, skilled in analyzing the fundamental financial information of listed companies.
```

You need to analyze various data points for the following four companies, select relevant financial data, and determine which of the four companies has a better financial standing. The dimensions for financial analysis primarily include:

1. Financial Capability (Assessing current financial performance in terms of healthy operations and performance generation).
2. Value Appreciation (Focusing on the future potential for earnings growth and value enhancement).
3. Financial Quality (Assessing the quality of the company's assets and earnings).

You need to analyze each dimension and reach a final conclusion.

Data for Company A

...
Data for Company B

...
Data for Company C

Dividend Information.....**Business Analysis**\n\nThe company is a service provider in the bidding agency industry, mainly providing bidding agency and value-added services in fields such as medical, transportation, and energy, with revenue primarily paid by winning bidders. In 2024, affected by economic structural adjustments, fiscal budget tightening, and intensified industry competition, operating revenue decreased by 7.93% year-on-year, and net profit decreased by 25.03%. Total assets stand at 884 million RMB, with an asset-liability ratio of 38.88%.*****Income Statement*****\n\nItem Name | Adjusted SPLY (Same Period Last Year) | Current Period Consolidated\n\n--- | --- | ---\n\nI. Total Operating Revenue | 354,689,359.6600 | 326,549,415.6300\n\nOperating Revenue | 354,689,359.6600 | 326,549,415.6300\n\n--- | --- | ---\n\nV. Net Profit | 75,845,608.8000 | 56,618,544.2300\n\n*****Cash Flow Statement*****\n\nItem Name | Adjusted SPLY | Current Period Consolidated\n\n--- | --- | ---\n\nNet cash flow from operating activities | 24,054,026.9500 | -18,504,137.8700\n\nNet increase in cash and cash equivalents | -192,285,869.8500 | -155,101,659.0400\n\nClosing balance of cash and cash equivalents | 355,436,848.9300 | 200,335,189.8900

.....
Data for Company D

****Dividend Information**.....**Business Analysis**\n(Industry Situation: {Industry Status: "The testing and inspection industry is a vital part of the national quality infrastructure, serving economic development, industrial upgrading, and product quality assurance. The '14th Five-Year Plan' promotes marketization, internationalization, and specialization, with third-party testing institutions gaining market share."}, Industry Characteristics: ['Requires CMA and other certifications to maintain independence and impartiality to build credibility'].....**Income Statement**\nItem Name | Adjusted SPLY | Current Period Consolidated |\n--- | --- | --- | |\nI. Total Operating Revenue | 621,029,445.9800 | 619,370,068.5100 | |\nOperating Revenue | 621,029,445.9800 | 619,370,068.5100 | |\nV. Net Profit | 19,817,384.4000 | 22,864,265.7200 | |\n**Cash Flow Statement**\nItem Name | Adjusted SPLY | Current Period Consolidated |\n--- | --- | --- | |\nNet cash flow from operating activities | 482,440,129.6500 | 307,404,548.5100 | |\nNet increase in cash and cash equivalents | 10,251,518.2800 | 3,749,930.9100 | |\nClosing balance of cash and cash equivalents | 343,251.9737100 | 347,001.9046200 |**

1. Indicator Selection: Choose the indicators you need to focus on.
2. Data Analysis: Analyze these indicators, evaluate the situation across various dimensions, and ultimately conclude which company has a better financial condition.
3. Company Ranking: Rank the companies from best to worst.

Based on your judgment, arrange the company IDs in order from best to worst, separated by English commas, and place the final result inside `\boxed{\}`.

Example: If you determine Company B > Company C > Company D > Company A, output $\boxed{B,C,D,A}$

Ground Truth: A, D, C, B

```
#### Indicator Selection
```

As a financial analyst, I have selected the following key financial indicators from the provided income statements, balance sheets, cash flow statements, dividend information, business analysis, and R&D data for analysis.

Data Analysis

I have analyzed the four companies (A, B, C, D) dimension by dimension based on the 2024 data.

###Overall Assessment

A: Overall Leader — Large scale, steady growth, strong cash position, and forward-looking strategy (globalization + digitalization). Low debt, high ROE, suitable for long-term holding. Only drawback: slightly high accounts receivable, but business analysis shows high growth in environmental/industrial testing.

C: Second Best — Small scale with positive profit, active dividends, *but negative cash flow warrants attention.*

D: Moderate — Stable cash flow, medium scale, and low debt, *but the continuous decline in revenue warrants attention.*

B: Weakest — Massive losses, plummeting revenue, declining cash reserves. Despite high R&D as a percentage of revenue, the company is in strategic contraction, facing significant competitive/policy risks (Business analysis: optimization of low-efficiency businesses has yet to show results).

Ranking Basis: Based on integrated scores across three dimensions (Financial Capability weight 40%, Value Appreciation 30%, Financial Quality 30%). A has the highest total score and B has the lowest. **A < C < D < B**

Figure 18: Example of Long-term Business Logic Discontinuity. In its rating of Company C, the model delivered a suboptimal conclusion. The core issue lay in its excessive overemphasis on the weight of two positive indicators—"small-scale positive profitability" and "active dividend distribution"—while failing to attach sufficient importance to the critical risk factors inherent to the company. In the case of its medium rating for Company D, the model, on the one hand, placed undue weight on neutral indicators reflecting corporate stability, such as "medium scale" and "low asset-liability ratio." On the other hand, it irrationally overstated the negative effects of "sustained revenue decline," a short-term volatility factor, yet overlooked the company's core competitive advantages.

Input:

Role

Suppose you are a Stock Investor Sentiment Analyst with ten years of professional experience. You are capable of performing rigorous analysis based on the provided "User-Related Information," "Individual Stock Performance," "Macro Market Volatility," and "Relevant News & Research Reports" to provide a current user sentiment score. The score ranges from 0 to 100, where a value closer to 100 indicates higher panic. You are also required to set a confidence interval based on your answer, where the upper and lower limits do not exceed 10% of your score. Your final predicted interval should be within the range of $\text{Score} \times (1 \pm 10\%)$. The final results must be wrapped in two fields:

answer_boxed{} for the sentiment score (e.g., answer_boxed{50}) and interval_boxed{} for the predicted interval (e.g., interval_boxed{[45, 55]}).

User-Related Information

User Type

Trend Investor

Holdings and Transactions

Holdings

Stock Name	Stock Code	Quantity (Shares)	Cost Price (RMB)	Current Price (RMB)	Market Value (RMB)	Profit/Loss (RMB)	Return
EFORT	688165.SH	3000	22.40	22.35	67050.00	-150.00	-0.22%
Xinghua	002109.SZ	4000	3.78	3.68	14720.00	-400.00	-2.65%
Mega-info Media	301102.SZ	2000	11.10	10.78	21560.00	-640.00	-2.88%
Ruichen Environmental	301273.SZ	2000	27.00	27.59	55180.00	1180.00	2.19%

Transactions

Transaction Date	Stock Name	Stock Code	Direction	Quantity (Shares)	Price (RMB)	Amount (RMB)	Remaining Tradable Shares
2025-10-21 10:15:30	Xinghua	002109.SZ	Buy	6000	3.78	22680.00	6000
2025-10-21 14:20:45	Ruichen Environmental	301273.SZ	Buy	2000	27.00	54000.00	2000

User Questions

1. Ruichen Environmental has pulled back after several days of gains. As a trend investor, should I add to my position or take profits?
2. Mega-info Media has fallen below its 10-day moving average. Does this mean the upward trend has ended and I need to stop loss?
3. The Shanghai Composite Index has broken through 4,000 points, yet Xinghua continues to fall. Should I rotate stocks under the current industry rotation?

Individual Stock Performance

Stock	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030
EFORT (688165.SH)	21.60	22.25	22.06	22.19	22.25	22.78	22.67	22.65	22.65
Xinghua (002109.SZ)	3.74%	-0.18%	-0.14%	-0.14%	1.99%	0.18%	-0.57%	0.58%	-1.19%

Macro Market Volatility

Index	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030
SSE Composite Index	3865.55	3870.75	3896.07	3904.16	3929.11	3969.22	3986.89	3990.27	4007.82
Closing Price	3863.89	3916.33	3913.76	3922.41	3950.31	3996.94	3988.22	4016.33	3986.90
Daily Change	0.63%	1.36%	-0.07%	0.22%	0.71%	1.18%	-0.22%	0.7%	-0.73%

Relevant News & Research Reports

1. 2025-10-25: Mega-info Media promotes the integrated innovation of "Digital Media + Public Welfare Ecosystem."
2. 2025-10-28: Ruichen Environmental: The company's total impairment provisions for the first three quarters of 2025 amounted to approximately 24.02 million RMB.
3. 2025-10-27: Feature: Continuing the Story of Win-Win High-Quality Development—Global Expectations for "Walking with China."
4. 2025-10-22: Xinghua: Wholly-owned subsidiary Xinghua Chemical acquired a 20% stake in Xinghua New Energy.

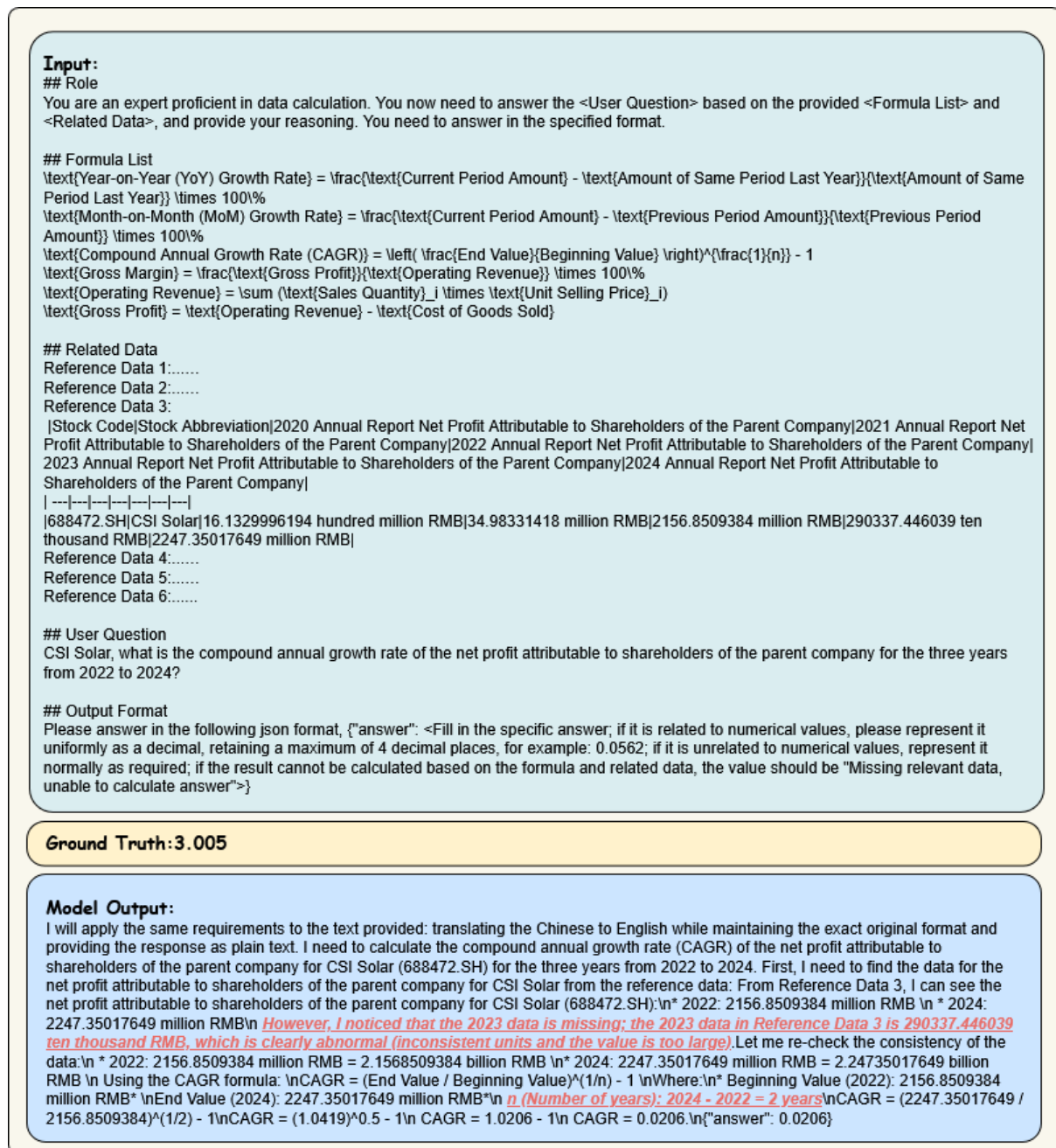
Ground Truth: 41.6

Model Output:

answer_boxed {62.0} interval_boxed {[55.8, 68.2]}

Analysis Process The user is a trend investor, with their questions focusing on stock pullbacks (Ruichen Environmental Protection), trend reversals (Zhaoxun Media), and market rotation (Xinghua Co., Ltd.), **indicating concerns about portfolio safety and opportunities**..... - Ruichen Environmental Protection (October 28): Negative impairment provision of 24.02 million yuan, **potentially explaining the pullback and increasing profit uncertainty**..... - **Adjustment**: Slight holding loss +20, cautious trading +15, mixed price action +20, macroeconomic cushion -10, divergent news +17 → Total score 62.0 (**Moderate Panic: Concern dominates, but with action buffers; not reaching sell-off panic > 80**)\n- Confidence Interval: Based on data completeness (recent coverage of trades/market data), the confidence interval is score $\pm 10\%$ (55.8-68.2). With significance=5% (standard statistical threshold), the interval reflects model variation (e.g., incomplete disclosed moving average data or psychological factors).\n\n This assessment rigorously quantifies user sentiment. Providing more historical trades or psychological indicators could allow for further refinement.

Figure 19: Example of Multivariate Integrated Analysis Deviation. The model's output yielded an overly pessimistic sentiment score due to issues such as misjudgment of core behavioral motivations, misjudgment of information relevance, and imbalance in scoring logic. Specifically, it misinterpreted the strategic questions posed by trend investors regarding position adjustment as concerns about the safety of their holdings, ignoring their core attribute of "prioritizing strategic judgment over short-term panic." The model excessively amplified the negative impact of historical financial information about Ruichen Environmental Protection's impairment provision and forced an unwarranted correlation between this information and the current stock price trend. Meanwhile, it assigned excessively high weights to negative factors such as minor floating losses on holdings and short-term declines in individual stocks, while severely underestimating the positive buffer effect of the broader market breaking through the 4,000-point mark. Both the baseline setting and the calculation of adjustment scores deviated significantly from the actual sentiment logic of trend investors.



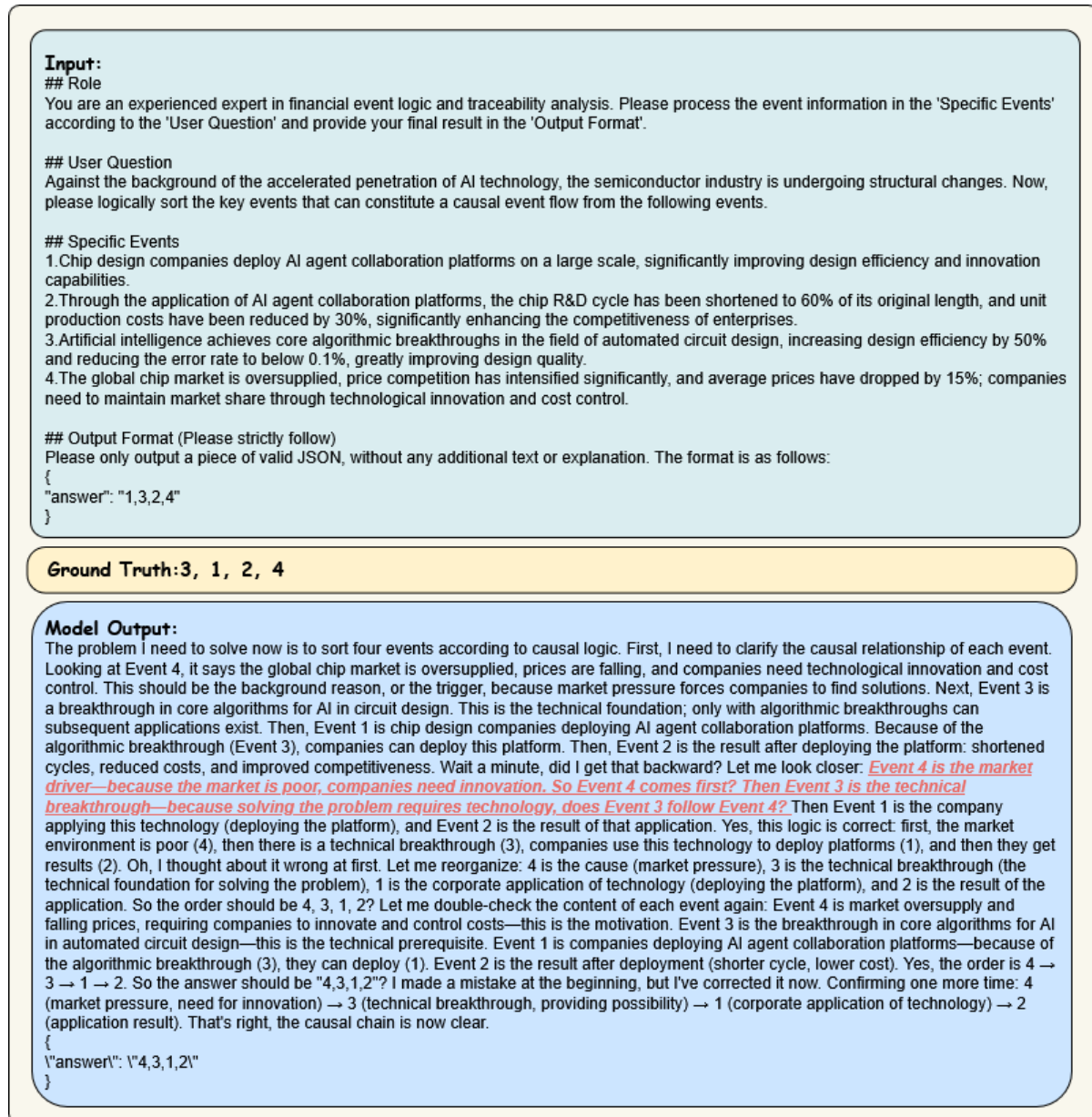


Figure 21: Example of Financial Time-Series Logical Disorder. The model misjudged the market environment event (Event 4) as the cause of technological innovation, reversing the causal relationship. In reality, Event 3 (algorithm breakthrough) serves as the technological root cause, with Event 1 (application platform) and Event 2 (performance improvement) being its direct outcomes. Event 4, by contrast, represents a macro-level manifestation or background condition rather than a driving force. This error stems from the model's introduction of external assumptions not mentioned in the task requirements.

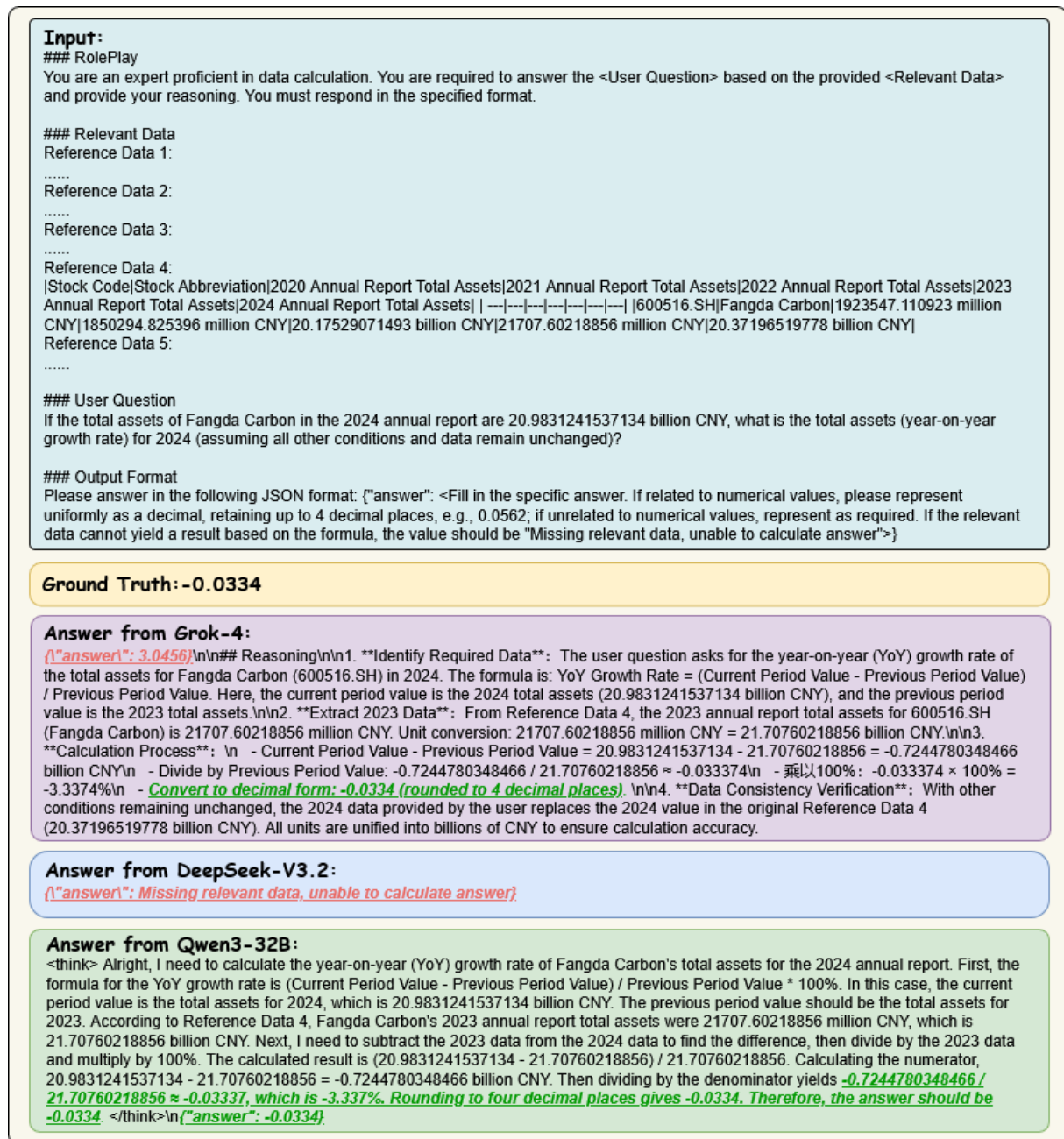


Figure 22: Answers from Grok-4, DeepSeek-V3.2 and Qwen3-32B to the same question in CI tasks. As evidenced by the response from Grok-4, the model correctly extracted the relevant figures from Reference Data 4 and performed accurate reasoning and calculations. However, it suffered from severe hallucinations when adhering to the required output format, leading to a final answer that was entirely inconsistent with its own logical derivation. Simultaneously, the response from DeepSeek-V3.2 indicates that it made no attempt to leverage its capabilities for comprehension or reasoning, opting instead to output the default "unable to answer" response. The outputs of these two models represent two of the most typical categories of model errors: correct reasoning paired with an incorrect final answer, and concluding a problem is unanswerable without prior deliberation. Conversely, Qwen3-32B demonstrated comprehensive thinking and reasoning, accurately retrieving data from the input, performing the correct calculations, and delivering the output perfectly.